① $e^{2t} + 3e^{-t} + 5$

$= L\{e^{2t}\} + L\{3e^{-t}\} + L\{5\}$

$= \frac{1}{s-2} + \frac{3}{s+1} + \frac{5}{s}$

② $\sinh^2 3t + \cosh^3 2t + (e^t + 5)^2$

$= \left[\frac{e^{3t} - e^{-3t}}{2}\right]^2 + \left[\frac{e^{2t} + e^{-2t}}{2}\right]^3 + (e^t)^2 + 5^2 + 10e^t$

$= \frac{1}{4}\left[(e^{3t})^2 + (e^{-3t})^2 - 2e^{3t}\cdot e^{-3t}\right] + \frac{1}{8}\left[(e^{2t})^3 + (e^{-2t})^3 + 3e^{2t}e^{-2t}(e^{2t} + e^{-2t})\right]$

$+ e^{2t} + 25 + 10e^t$

$\Rightarrow \frac{1}{4}\left[e^{6t} + e^{-6t} - 2\right] + \frac{1}{8}\left[e^{6t} + e^{-6t} + 3e^{2t} + 3e^{-2t}\right]$

$+ e^{2t} + 25 + 10e^t$

$= \frac{1}{4}\left[\frac{1}{s-6} + \frac{1}{s+6} - \frac{2}{s}\right] + \frac{1}{8}\left[\frac{1}{s-6} + \frac{1}{s+6} + \frac{3}{s-2} + \frac{3}{s+2}\right] + \frac{1}{s-2} + \frac{25}{s}$

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UNIT-3
LAPLACE TRANSFORMS

Let $f(t)$ be a real valued function for $t \geq 0$ and s is a parameter ($s > 0$) which is real or complex then integral from 0 to ∞

$$\boxed{L\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt}$$

Provided integral should exist and finite.

⇒ Laplace Transforms of standard function:-

① $f(t) = a$

$$L\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt = \int_0^\infty e^{-st}\, a\,dt$$

$$= a\left[\frac{e^{-st}}{-s}\right]_0^\infty = \frac{-a}{s}\left[e^{-s(\infty)} - e^{-s(0)}\right]$$

$$= a/s$$

② $f(t) = e^{at}$

$$L\{f(t)\} = \int_0^\infty e^{-st} \times e^{at}\,dt = \int_0^\infty e^{(s+a)t}\,dt$$

$$= \left[\frac{e^{(-s+a)t}}{(-s+a)}\right]_0^\infty$$

$$= \frac{1}{(-s+a)}\left[e^{\infty} - 1\right] = \infty$$

WRONG

$$= \int_0^\infty e^{-(s-a)t}\,dt = \left[\frac{e^{-(s-a)t}}{-(s-a)}\right]_0^\infty$$

$$= \frac{1}{-(s-a)}\left[e^{-\infty} - e^{0}\right]$$

$$= \frac{1}{(s-a)}$$

$\Rightarrow f(t) = \sin at$

$$L\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt = \int_0^\infty e^{-st} \times \sin at\,dt$$

$a = -s,\ b = a,\ x = t$

$$= \left\{e^{-st}\left[\frac{-s\sin ax - a\cos ax}{s^2+a^2}\right]\right\}_0^\infty$$

$$= \frac{0 - (-a)}{s^2+a^2}$$

$$= \frac{a}{s^2+a^2}$$

① $\int e^{ax}\sin bx\,dx = e^{ax}\left[\frac{a\sin bx - b\cos bx}{a^2+b^2}\right]$

② $\int e^{ax}\cos bx\,dx = e^{ax}\left[\frac{a\cos bx + b\sin bx}{a^2+b^2}\right]$

NOTES!

① $L\{k\} = \frac{k}{s}$; $s > 0$

② $L\{e^{at}\} = \frac{1}{s-a}$; $s > a$

③ $L\{e^{-at}\} = \frac{1}{s+a}$; $s > a$

④ $L\{\sinh at\} = \frac{e^{at} - e^{-at}}{2} = \frac{a}{s^2-a^2}$

$\because\ s^2 - a^2 > 0$

$s^2 > \pm a^2$

$s > |a|$

⑤ L{cos

⇒ Regions

⑥ L{s

⑦ L{c

⇒ L{

L{t

Sol:

Put

s

③ $\Gamma(n) = \frac{\Gamma(n+1)}{n}$ ∵ n is -ve fraction.

④ Imp:
① 0 is not defined
② Also gamma function is not defined for -ve integer.

⑤ $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

Q Find the Laplace transform of the following functions

① $(e^{2t} + e^{-3t})^2$

Sol: $(a+b)^2 = a^2 + 2ab + b^2$

$\Rightarrow (e^{2t})^2 + (e^{-3t})^2 + 2e^{2t}e^{-3t}$

$\Rightarrow e^{4t} + e^{-6t} + 2e^{-t}$

$\Rightarrow \frac{1}{(s-4)} + \frac{1}{(s+6)} + \frac{2}{(s+1)}$

② $(1+e^t)^2$

Sol: $1^2 + (e^t)^2 + 2e^t$

$\Rightarrow 1 + e^{2t} + 2e^t$

$\Rightarrow \frac{1}{s} + \frac{1}{(s-2)} + \frac{2}{(s-1)}$

③ $\sin 5t \cos 7t$

$\Rightarrow \sin A \times \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$

Sol $\sin 5t \cos 7t = \frac{1}{2}\left[\sin 12t - \sin 2t\right]$

$= \frac{1}{2}\left[\frac{12}{s^2+144} - \frac{2}{s^2+4}\right]$

$= \frac{6}{s^2+144} - \frac{1}{s^2+4}$

(4) $\sin 3t \times \sin 2t$

$\Rightarrow$

$\because \sin A \sin B = \frac{-1}{2}\left[\cos(A+B) - \cos(A-B)\right]$

$\because \cos A \cos B = \frac{-1}{2}\left[\cos(A+B) + \cos(A-B)\right]$

$\Rightarrow \sin 3t \sin 2t = \frac{-1}{2}\left[\cos 5t - \cos t\right]$

$= \frac{-1}{2}\left[\frac{s}{s^2+25} - \frac{s}{s^2+1}\right]$

(5) $\cos 2t \cos 7t$

Sol $\cos 2t \cdot \cos 7t = \frac{+1}{2}\left[\cos 9t + \cos 5t\right]$

$= \frac{+1}{2}\left[\frac{s}{s^2+81} + \frac{s}{s^2+25}\right]$

(6) $(1+t^3)^2$

Sol $1^2 + (t^3)^2 + 2t^3$

$\Rightarrow 1 + t^6 + 2t^3$

$\Rightarrow \frac{1}{s} + \frac{6!}{s^7} + \frac{2 \times 3!}{s^4} \Rightarrow \frac{1}{s} + \frac{720}{s^7} + \frac{12}{s^4}$

(11) $L\{2^{-t}\}$

Sol:
$2^{-t} = e^{\log 2^{-t}}$ $(\because s > \log 2)$
$= e^{-t\log 2}$
$= \frac{1}{s+\log 2}$

$e^{\log a^{-t}}$
$\therefore L\{e^{-t\log a}\}$
$= \frac{1}{(s+\log a)}$

(12) $L\{(1-K)^t\}$

Sol: $(1-K)^t = e^{\log(1-K)^t}$
$= e^{t\log(1-K)}$
$= \frac{1}{s-\log(1-K)}$

(13) $L\left\{\left(\sqrt{t}+\frac{1}{\sqrt{t}}\right)^3\right\}$

Sol: $\left(\sqrt{t}+\frac{1}{\sqrt{t}}\right)^3 = (\sqrt{t})^3 + \frac{1}{(\sqrt{t})^3} + 3\sqrt{t}\times\frac{1}{\sqrt{t}}\left(\sqrt{t}+\frac{1}{\sqrt{t}}\right)$
$= t^{3/2} + t^{-3/2} + 3\left(\sqrt{t}+\frac{1}{\sqrt{t}}\right)$

$= \frac{3/4\sqrt{\pi}}{s^{5/2}} + \frac{1\sqrt{\pi}}{-3/4\, s^{-1/2}} + \frac{3\frac{1}{2}\sqrt{\pi}}{s^{3/2}} + \frac{3\sqrt{\pi}}{s^{1/2}}$

$= \frac{3\sqrt{\pi}}{4s^{5/2}} + \frac{4\sqrt{\pi}}{3s^{1/2}}$

$= \frac{3\sqrt{\pi}}{4s^{5/2}} + \frac{-2\sqrt{\pi}}{s^{-1/2}} + \frac{3\sqrt{\pi}}{2s^{3/2}} + \frac{3\sqrt{\pi}}{\sqrt{s}}$

(14) $\sinh at + \cosh^2 3t$

Sol: $\frac{a}{s^2-a^2}$, $\cosh^2 3t = \left[\frac{e^{3t}+e^{-3t}}{2}\right]^2$

$+\frac{1}{4}\left[\frac{1}{s-6} + \frac{1}{s+6} + \frac{2}{s}\right] = \frac{1}{4}\left[e^{6t}+e^{-6t}+2\right]$

(15) $\sin^3 4t$

Sol: $\sin^3\theta = \frac{3\sin\theta - \sin 3\theta}{4}$

$\sin^3 4t = \left\{\frac{3\sin 4t - \sin 12t}{4}\right\}$

$= \frac{1}{4}\left\{\frac{3}{s^2+16} - \frac{12}{s^2+144}\right\}$

(16) $\cos^3 8t$

Sol: $\cos^3\theta = \frac{\cos 3\theta + 3\cos\theta}{4}$

$\cos^3 8t = \left\{\frac{\cos 24t + 3\cos 8t}{4}\right\}$

$= \frac{1}{4}\left\{\frac{s}{s^2+24^2} + \frac{3s}{s^2+64}\right\}$

$= \frac{1}{4}\left\{\frac{s}{s^2+576} + \frac{3s}{s^2+64}\right\}$

$L\{f(t)\} = \int_0^\infty e^{-st} f\{t\}\,dt = F(s)$

(7) $e^{-t}t^7$

Sol: $f(t) = t^7$

$L\{f(t)\} = \frac{\lfloor(7+1)}{s^{7+1}} = \frac{\lfloor 7!}{s^8} = \frac{504}{s^8}$

$\therefore L\{e^{-t}f(t)\} = \frac{504}{(s+1)^8}$

(8) $\cosh^2 3t \sin t$

Sol: $\cosh^2 3t = \left[\frac{e^{3t}+e^{-3t}}{2}\right]^2 = \frac{1}{4}\left[e^{6t}+e^{-6t}+2\right]$

Let $f(t) = \sin t$

$L\{f(t)\} = \frac{1}{s^2+1}$

$L\{\cosh^2 3t \sin t\} = \frac{1}{4}L\{(e^{6t}+e^{-6t}+2)\sin t\}$

$= \frac{1}{4}\left[\frac{1}{(s-6)^2+1} + \frac{1}{(s+6)^2+1} + \frac{2}{s^2+1}\right]$

(9) $e^{6t}\sin^3 5t$

Sol: $f(t) = \sin^3 5t = \frac{3\sin 5t - \sin 15t}{4}$

$L\{f(t)\} = \frac{1}{4}\left[\frac{3(5)}{s^2+25} - \frac{15}{s^2+225}\right] = \frac{15}{4}\left[\frac{1}{s^2+25} - \frac{1}{s^2+225}\right]$

$L\{e^{6t}f(t)\} = \frac{15}{4}\left[\frac{1}{(s-6)^2+25} - \frac{1}{(s-6)^2+225}\right]$

$$= \frac{1}{4}\left[\left(\frac{1}{(s-3)}\right) + \left(\frac{s-3}{(s-3)^2+36}\right) + \frac{1}{(s+3)} + \frac{s+3}{(s+3)^2+36}\right]$$

(4) $e^{-7t}\cos^3 8t$

Sol: $f(t) = \cos^3 8t = \frac{3\cos 8t + \cos 24t}{4}$

$$L\{f(t)\} = \frac{1}{4}\left[3.\frac{s}{s^2+64} + \frac{s}{s^2+576}\right] = \frac{s}{4}\left[\frac{3}{s^2+64} + \frac{1}{s^2+576}\right]$$

(5) $e^{-2t}t^{-11/2}$

Sol: ~~$f(t) = \cos^3 8t = \frac{3\cos 8t + \cos 24t}{4}$~~

$$f(t) = t^{-11/2}$$

$$L\{f(t)\} = \frac{\Gamma(-11/2+1)}{s^{-11/2+1}} = \frac{\Gamma(-9/2)}{s^{-9/2}} = \frac{\Gamma(-1/2+1)}{(-9/2)(-7/2)(-5/2)(-1/2)s^{-9/2}}$$

$$= \frac{\Gamma(1/2)}{\frac{-945}{32}s^{-9/2}} = \frac{-32\sqrt{\pi}}{945 s^{-9/2}}$$

$$\therefore L\{e^{-2t}f(t)\} = \frac{-32\sqrt{\pi}}{945(s+2)^{-9/2}}$$

(6) $e^{6t}t^{-11/4}$

Sol: $f(t) = t^{-11/4}$

$$L\{f(t)\} = \frac{\Gamma(-11/4+1)}{s^{-11/4+1}} = \frac{\Gamma(-7/4)}{s^{-7/4}} = \frac{\Gamma(-3/4+1)}{(-7/4)(-3/4)s^{-7/4}} = \frac{16\Gamma(1/4)}{21 s^{-7/4}}$$

$$\therefore L\{e^{6t}f(t)\} = \frac{16\Gamma(1/4)}{21(s-6)^{-7/4}}$$

③ $L\{f(2t/3)\} = \frac{3}{2}F(\frac{3s}{2})$

=

Q. Find the Laplace transform of the functions:

① $e^{-t}t^{7/2}$

Sol: Let $f(t) = t^{7/2}$

$= \frac{\Gamma(7/2+1)}{s^{7/2+1}} = \frac{(7/2)(5/2)(3/2)[(1/2)}{s^{9/2}} = \frac{105\sqrt{\pi}}{16\, s^{9/2}}$

$\therefore L\{e^{-t}t^{7/2}\} = \frac{105\sqrt{\pi}}{16(s+1)^{9/2}}$

② $e^{6t}\sin^2 2t$

Sol: $f(t) = \sin^2 2t = \frac{1-\cos 4t}{2}$

$L\{F(t)\} = \frac{1}{2}\left[\frac{1}{s} - \frac{s}{s^2+16}\right]$

$L\{e^{6t}\sin^2 2t\} = \frac{1}{2}\left[\frac{1}{(s-6)} - \frac{(s-6)}{(s-6)^2+16}\right]$

③ $\cosh 3t \cos^2 3t$

Sol: $\cosh 3t = \frac{e^{3t}+e^{-3t}}{2}$; $\cos^2 3t = \frac{1+\cos 6t}{2} = f(t)$

$L\{f(t)\} = \frac{1}{2}\left[\frac{1}{s} + \frac{s}{s^2+36}\right]$

$\therefore L\left\{\frac{e^{3t}\cos^2 3t}{2} + \frac{e^{-3t}\cos^2 3t}{2}\right\} =$

$= \frac{1}{2}\left[\frac{1}{2}\left(\frac{1}{s-3} + \frac{(s-3)}{(s-3)^2+36}\right)\right] + \frac{1}{2}\left[\frac{1}{2}\left(\frac{1}{s+3} + \frac{(s+3)}{(s+3)^2+36}\right)\right]$

(10) $\sinh t\, t^{5/2}$

Sol $\sinh t = \frac{e^t - e^{-t}}{2}$

Let $t^{5/2} = f(t)$

$$L\{f(t)\} = \frac{\Gamma(\frac{5}{2}+1)}{s^{\frac{5}{2}+1}} = \frac{(\frac{5}{2})(\frac{3}{2})(\frac{1}{2})\Gamma(\frac{1}{2})}{s^{7/2}} = \frac{15\sqrt{\pi}}{8s^{7/2}}$$

$$L\{\sinh t\, t^{5/2}\} = \frac{1}{2}\left[L\{e^t - e^{-t}(t^{5/2})\}\right] = \frac{1}{2}\left[\frac{15\sqrt{\pi}}{8(s-1)^{7/2}} - \frac{15\sqrt{\pi}}{8(s+1)^{7/2}}\right]$$

$$= \frac{15\sqrt{\pi}}{16}\left[\frac{1}{(s-1)^{7/2}} - \frac{1}{(s+1)^{7/2}}\right]$$

(11) $2^t + 10$

Sol $2^t = e^{\log 2^t} = e^{t\log 2}$

$f(t) = t^{10}$

$$L\{f(t)\} = \frac{\Gamma(10+1)}{s^{10+1}} = \frac{10!}{s^{11}}$$

$$L\{2^t + 10\} = \frac{10!}{(s+\log 2)^{11}}$$

⇒ Multiplication By t^n in t domain of differentiation in S domain

$$L\{t^n f(t)\} = (-1)^n \frac{d^n F(s)}{ds^n}$$

$$= \left[\frac{18s^2 - 54}{(s^2+9)^3}\right]$$

$$L\{t^2 e^{2t}\sin 3t\} = \left[\frac{18(s-2)^2 - 54}{((s-2)^2+9)^3}\right]$$

Q $L\{t^{10}e^{3t}\}$

Sol: Consider, $f(t) = t^{10}$

$$L\{f(t)\} = \frac{10!}{s^{10+1}} = \frac{10!}{s^{11}}$$

$$\therefore L\{t^{10}e^{3t}\} = \frac{10!}{(s-3)^{11}}$$

Q $\{t^3 \cosh^2 4t\}$

Sol: Consider, $f(t) = t^3$

$$L\{f(t)\} = \frac{3!}{s^{3+1}} = \frac{3!}{s^4}$$

$$\because \cosh^2 4t = \left[\frac{e^{4t}+e^{-4t}}{2}\right]^2 = \frac{1}{4}\left[e^{8t}+e^{-8t}+2\right]$$

$$L\{\cosh^2 4t \cdot t^3\} = L\left\{\left(\frac{e^{8t}+e^{-8t}+2}{4}\right)\cdot\frac{3!}{s^4}\right\}$$

$$= \frac{3!}{4(s-8)^4} + \frac{3!}{4(s+8)^4} + \frac{8\cdot 3!}{4(s)}$$

→ Laplace transform of $\frac{f(t)}{t}$:—

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s)\,ds$$

If $L\{f(t)\} = F(s)$, then $\boxed{L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s)\,ds}$

Important Formulae :—

① $\int \frac{1}{s^2+a^2}\,ds = \frac{1}{a}\tan^{-1}\left(\frac{s}{a}\right)$

② $\int \frac{s}{s^2+a^2}\,ds = \int \frac{1}{t}\,\frac{dt}{2}$

$= \frac{1}{2}\log(s^2+a^2)$

$s^2+a^2 = t$

Diff w.r.t 's'

$2s\,ds = dt$

$s\,ds = \frac{dt}{2}$

③ $\int \frac{s}{s^2-a^2}\,ds = \frac{1}{2}\log(s^2-a^2)$

④ $\sin A\cos B = \frac{1}{2}\left[\sin(A+B)+\sin(A-B)\right]$

⑤ $\cos A\cos B = \frac{1}{2}\left[\cos(A+B)+\cos(A-B)\right]$

⑥ $\sin A\sin B = -\frac{1}{2}\left[\cos(A+B)-\cos(A-B)\right]$

④ $\int \frac{1}{(s-a)}\,ds = \log(s-a)$

Now,

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s)\,ds$$

$$L\left\{\frac{\sin^3 t}{t}\right\} = \int_s^\infty \frac{3}{4}\left[\frac{1}{s^2+1} - \frac{1}{s^2+9}\right]$$

$$= \frac{3}{4}\left[\tan^{-1}(s) - \frac{1}{3}\tan^{-1}\left(\frac{s}{3}\right)\right]_s^\infty$$

$$= \frac{3}{4}\left[\frac{\pi}{2} - \tan^{-1}(s)\right]$$

$$= \frac{1}{2}\left[\frac{\pi}{2} - \tan^{-1}(s)\right] = \frac{\cot^{-1}(s)}{2}$$

Now $L\left\{\frac{e^{2t}\sin^3 t}{t}\right\} = \frac{\cot^{-1}(s-2)}{2}$.

$$= \frac{3}{4}\left[\frac{\pi}{2} - \tan^{-1}s - \frac{1}{3}\left(\frac{\pi}{2} - \tan^{-1}\left(\frac{s}{3}\right)\right)\right]$$

$$= \frac{3}{4}\left[\cot^{-1}s - \frac{1}{3}\cot^{-1}\left(\frac{s}{3}\right)\right]$$

Now, $L\left\{\frac{e^{2t}\sin^3 t}{t}\right\} = \frac{3}{4}\left[\cot^{-1}(s-2) - \frac{1}{3}\cot^{-1}\left(\frac{s-2}{3}\right)\right]$

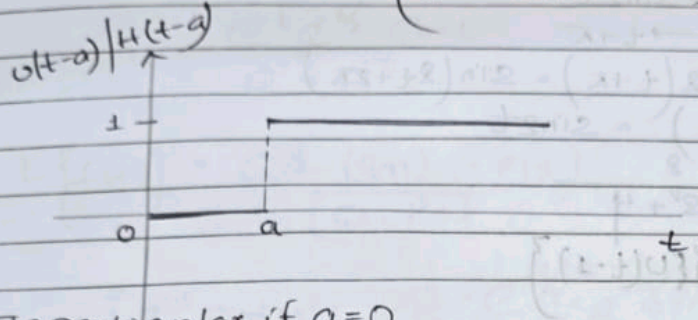
UNIT STEP FUNCTION / HEAVISIDE FUNCTION

$$u(t-a) / H(t-a) = \begin{cases} 0 & t<a \\ 1 & t>a \end{cases}$$

Unit step-function / Heaviside function is denoted by $u(t-a)$

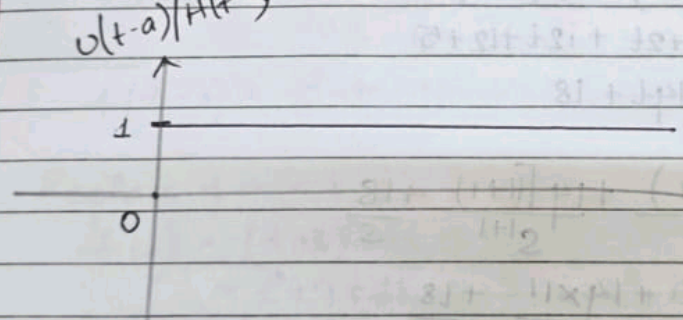
~~$H(t-a)$~~ =

$$u(t-a) / H(t-a) = \begin{cases} 0 & t<a \quad \text{(off)} \\ 1 & t>a \quad \text{(on)} \end{cases}$$

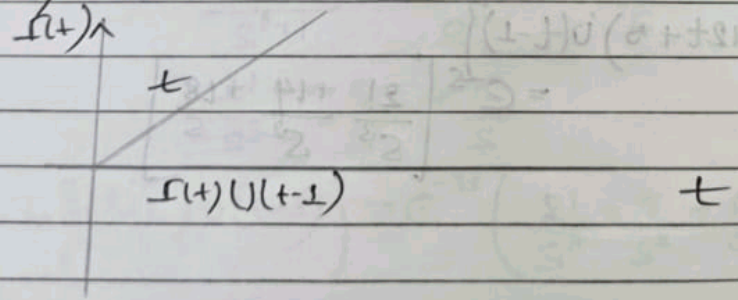


⇒ In particular if $a=0$

$$u(t) = \begin{cases} 0 & t<0 \\ 1 & t>0 \end{cases}$$



⇒ $f(t) = t$



Apply Laplace Transform to Both sides

$$L\{f(t)\} = L\{t\} + L\{(t^2-t)\,u(t-2)\} \quad \text{---(1)}$$

$$= \frac{1}{s^2} + e^{-2s}\left[\frac{2}{s^3} + \frac{3}{s^2} + \frac{2}{s}\right]$$

$f(t) = t^2 - t$

Replace $t \to t+2$

$f(t) = (t+2)^2 - (t+2)$

$= t^2 + 4 + 4t - t - 2$

$f(t) = t^2 + 3t + 2$

$$L\{f(t)\} = L\{t^2+3t+2\}$$

$$= \frac{2}{s^3} + \frac{3}{s^2} + \frac{2}{s}$$

$= F(s)$

(2) $$f(t) = \begin{cases} e^t = f_1(t) & 0 < t < 1 \\ 1 = f_2(t) & t > 1 \end{cases}$$

Sol:

$$f(t) = f_1(t) + [f_2(t) - f_1(t)]\,u(t-a)$$

$$f(t) = e^t + [1 - e^t]\,u(t-1)$$

$$f(t) = e^t + [(1-e^t)\,u(t-1)]$$

Apply Laplace Transform to Both sides

$$L\{f(t)\} = L\{e^t\} + L\{(1-e^t)\,u(t-1)\}$$

$$= \frac{1}{(s-1)} + e^{-s}\left[\frac{1}{s} - \frac{e}{(s-1)}\right]$$

(21)

(4) $f(t) = \begin{cases} 1 & 0 < t \leq 1 \\ 2t & 1 < t \leq 2 \\ 3t^2 & t > 2 \end{cases}$

Sol:

$f(t) = f_1(t) + [f_2(t) - f_1(t)]u(t-a) + [f_3(t) - f_2(t)]u(t-b)$

$f(t) = 1 + [2t - \text{[illegible]}]u(t-1) + [3t^2 - 2t]u(t-2)$

$f(t) = 1 + [(2t-1)u(t-1)] + [(3t^2-2t)u(t-2)]$

Apply LT on Both sides

$L\{f(t)\} = L\{1\} + L\{(2t-1)u(t-1)\} + L\{(3t^2-2t)u(t-2)\}$

(1) (2)

$= \frac{1}{s} + e^{-s}\left[\frac{2}{s^2} + \frac{1}{s}\right] + e^{-2s}\left[\frac{6}{s^3} + \frac{8}{s} + \frac{10}{s^2}\right]$

Now, (1) ⇒

$f(t-1) = 2t - 1$

Replace $t \to t+1$

$f(t) = 2(t+1) - 1$

$f(t) = 2t + 1$

$L\{f(t)\} = \frac{2}{s^2} + \frac{1}{s}$

Now (2) ⇒

$f(t-2) = 3t^2 - 2t$

Replace $t \to t+2$

$f(t) = 3(t+2)^2 - 2(t+2)$

$= 3[t^2 + 4 + 4t] - 2t - 4$

$= 3t^2 + 12 + 12t - 2t - 4$

$f(t) = 3t^2 + 8 + 10t$

$L\{f(t)\} = \frac{3 \times 2}{s^3} + \frac{8}{s} + \frac{10}{s^2}$

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(1) $f(t) = \begin{cases} \sin t & 0 < t < \pi \\ \sin 2t & \pi < t < 2\pi \\ \sin 3t & t > 2\pi \end{cases}$

Sol: $f(t) = f_1(t) + [f_2(t) - f_1(t)]\,U(t-a) + [f_3(t) - f_2(t)]\,U(t-b)$

$f(t) = \sin t + [\sin 2t - \sin t]\,U(t-\pi) + [\sin 3t - \sin 2t]\,U(t-2\pi)$

Apply L.T on Both sides —

$L\{f(t)\} = L\{\sin t\} + L\{(\sin 2t - \sin t)\,U(t-\pi)\} + L\{(\sin 3t - \sin 2t)\,U(t-2\pi)\}$

(1) (2) (3)

$L\{f(t)\} = \frac{1}{s^2+1} + e^{-\pi s}\left[\frac{2}{s^2+4} - \frac{1}{s^2+1}\right] + e^{-2\pi s}\left[\frac{3}{s^2+9} - \frac{2}{s^2+4}\right]$

Now, (1) ⇒

$f(t-\pi) = \sin 2t - \sin t$

Replace $t \to t+\pi$

$f(t) = \sin(2t+2\pi) - \sin(t+\pi)$

$f(t) = \sin(2t+2\pi) - \sin(t+\pi)$

$f(t) = \sin 2t + \sin t$

$L\{f(t)\} = \frac{2}{s^2+4} + \frac{1}{s^2+1}$

Now, (2) ⇒

$f(t-2\pi) = \sin 3t - \sin 2t$

Replace $t \to t+2\pi$

$f(t) = \sin 3(t+2\pi) - \sin 2(t+2\pi)$

$f(t) = \sin(3t+6\pi) - \sin(2t+4\pi)$

$f(t) = \sin 3t - \sin 2t$

$L\{f(t)\} = \frac{3}{s^2+9} - \frac{2}{s^2+4}$

(c) $L\left\{\frac{2\delta(t-3)}{t} + \frac{3\delta(t-2)}{t}\right\}$

$e^{-3s}\left(\frac{3}{3}\right) + e^{-2s}\left(\frac{3}{2}\right)$

(d)

NOTE:

$L\{U(t)\} = \frac{1}{s}$

$\Rightarrow \frac{2t^{1/2}}{\frac{1}{2}\sqrt{\pi}\sqrt{\pi}} - 1 + \frac{2t^{3/2}}{\frac{3}{2}\frac{1}{2}\sqrt{\pi}} - t$

$\Rightarrow \frac{4t^{1/2}}{\sqrt{\pi}} - 1 \frac{8t^{3/2}}{3\sqrt{\pi}} - t$

(3) $\frac{3s-8}{4s^2+25}$

$\Rightarrow \frac{3s}{4s^2+25} - \frac{8}{4s^2+25}$

$\Rightarrow \frac{3s}{4\left(s^2+\frac{25}{4}\right)} - \frac{8}{4\left(s^2+\frac{25}{4}\right)}$

$\Rightarrow \frac{3}{4}\cos\frac{5}{2}t - \frac{8}{4}\cos\frac{5}{2}t$

$\Rightarrow \frac{3}{4}\cos\frac{5}{2}t - 2\cos\frac{5}{2}t$

$\Rightarrow \frac{3}{4}\cos\frac{5}{2}t - 2 \times 1 \sin\left(\frac{5}{2}t\right)$

(4) $\frac{5s+10}{9s^2-16}$

$\Rightarrow \frac{5s}{9s^2-16} + \frac{10}{9s^2-16}$

$\Rightarrow \frac{5s}{9\left(s^2-16/9\right)} + \frac{10}{9\left(s^2-16/9\right)}$

$\Rightarrow \frac{5s}{9\left(s^2-(4/3)^2\right)} + \frac{10}{9\left(s^2-(4/3)^2\right)}$

$\Rightarrow \frac{5}{9}\cosh\left(\frac{4}{3}t\right) + \frac{10}{9} \times \frac{1}{3} \sinh\left(\frac{4}{3}t\right)$

$\Rightarrow \frac{5}{9}\cosh\left(\frac{4}{3}t\right) + \frac{5}{6}\sinh\left(\frac{4}{3}t\right)$

NOTE:

(1) $L^{-1}\left\{\frac{1}{(s-a)^2+a^2}\right\} = \frac{1}{a}e^{at}\sin at$

(2) $L^{-1}\left\{\frac{(s-a)}{(s-a)^2+a^2}\right\} = e^{at}\cos at$

(3) $L^{-1}\left\{\frac{s}{(s^2+1)^2}\right\} = \frac{1}{2}t\sin t.$

(4) $L^{-1}\left\{\frac{s^2-a^2}{(s^2+1)^2}\right\} = t\cos at$

② $\frac{1}{s^2+6s+18}$

Sol: Completing the square for the denominator —

$s^2+6s+18$

$\Rightarrow \left[(s+\frac{6}{2})^2+18-(3)^2\right]$

$\Rightarrow (s+3)^2+3^2$

Now, $\frac{1}{(s+3)^2+3^2}$

Change $s+3$ to s

$\Rightarrow \frac{1}{s^2+3^2} e^{-3t}$

$\Rightarrow e^{-3t} L^{-1}\left\{\frac{1}{s^2+3^2}\right\} \Rightarrow e^{-3t}\frac{1}{3}\sin 3t$ =

③ $\frac{7s+4}{4s^2+4s+9}$

Sol: Completing the square for the denominator —

$4s^2+4s+9$

$\Rightarrow 4\left[s^2+s+\frac{9}{4}\right]$

$\Rightarrow 4\left[(s+\frac{1}{2})^2+\frac{9}{4}-(\frac{1}{2})^2\right]$

$\Rightarrow 4\left[(s+\frac{1}{2})^2+\frac{8}{4}\right] \Rightarrow 4\left[(s+\frac{1}{2})^2+(\sqrt{2})^2\right]$

Now, Numerator —

$7(s+\frac{1}{2}-\frac{1}{2})+4 \Rightarrow 7(s+\frac{1}{2})-\frac{7}{2}+4$

$\Rightarrow 7(s+\frac{1}{2})+\frac{1}{2}$

METHOD OF PARTIAL FRACTION

Q Find the L^{-1} of the following functions:-

① $\frac{s}{(s-2)(s-1)}$

Sol: $\frac{s}{(s-2)(s-1)} = \frac{A}{(s-2)} + \frac{B}{(s-1)}$ ——— (1)

Consider,

$\frac{s}{(s-2)(s-1)} \Big| = \frac{A}{(s-2)} \Rightarrow \frac{2}{1} = A$

$A = 2$

$s = 2$

$\frac{s}{(s-2)(s-1)} \Big| = \frac{B}{(s-1)} \Rightarrow \frac{1}{-1} = B$

$B = -1$

$s = 1$

(1) $\Rightarrow \frac{s}{(s-2)(s-1)} = \frac{2}{(s-2)} - \frac{1}{(s-1)}$ (Find Inverse Laplace)

$= 2e^{2t} - e^{t}$ → Ye ladki bhot ag jageg!!

YELAHANKA

② $\frac{s+2}{(s+1)^2(s-2)}$

Sol: $\frac{s+2}{(s+1)^2(s-2)} = \frac{A}{(s+1)} + \frac{B}{(s+1)^2} + \frac{C}{(s-2)}$ ——— (1)

$\times (s+1)^2(s-2)$

$\Rightarrow (s+2) = A(s+1)(s-2) + B(s-2) + C(s+1)^2$ ——— (2)

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(5) $\frac{s+6}{(2s+1)(2s+3)}$

Sol $\frac{s+6}{(2s+1)(2s+3)} = \frac{A}{(2s+1)} + \frac{B}{(2s+3)} \longrightarrow (1)$

Compare, $\frac{(s+6)}{(2s+1)(2s+3)} = \frac{A}{(2s+1)}$

$s = -\frac{1}{2}$

$\Rightarrow \frac{-\frac{1}{2}+6}{2(-\frac{1}{2})+3} = A$

$\Rightarrow \frac{11/2}{2} = A$

$A = \frac{11}{4}$

Compare, $\frac{s+6}{(2s+1)(2s+3)} = \frac{B}{(2s+3)}$

$s = -\frac{3}{2}$

$\Rightarrow \frac{-3/2+6}{2(-3/2)+1} = B$

$B = -\frac{9}{4}$

(1) $\Rightarrow$

$$\frac{s+6}{(2s+1)(2s+3)} = \frac{11}{4}\left(\frac{1}{2(s+1/2)}\right) - \frac{9}{4}\left(\frac{1}{2(s+3/2)}\right)$$

$$= \frac{11}{8}e^{-\frac{1}{2}t} - \frac{9}{8}e^{-\frac{3}{2}t}$$

(6) $\frac{s+5}{s^2(s-1)(s+2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{(s-1)} + \frac{D}{(s+2)} \longrightarrow (1)$

Sol Multiply throughout by $\times s^2(s-1)(s+2)$

$(s+5) = As(s-1)(s+2) + B(s-1)(s+2) + C(s+2)s^2 + Ds^2(s-1)$

Consider, $\frac{s+5}{s^2(s-1)(s+2)} = \frac{B}{s^2}$

$s = 0$

$\Rightarrow \frac{0+5}{(0-1)(0+2)} = B$

$B = -\frac{5}{2}$

(12) $\frac{s^2+s+2}{(s-2)^6}$

Sol:

$$\frac{(s-2+2)^2+(s-2+2)+2}{(s-2)^6}$$

$$\Rightarrow \frac{(s-2)^2+4+2\times2(s-2)+(s-2)+4}{(s-2)^6}$$

$$\Rightarrow \frac{(s-2)^2+4(s-2)+(s-2)+8}{(s-2)^6}$$

$$\Rightarrow \frac{(s-2)^2+5(s-2)+8}{(s-2)^6}$$

$$\Rightarrow \frac{(s-2)^2}{(s-2)^6}+\frac{5(s-2)}{(s-2)^6}+\frac{8}{(s-2)^6}$$

$$\Rightarrow \frac{1}{(s-2)^4}+\frac{5}{(s-2)^5}+\frac{8}{(s-2)^6}$$

$$\Rightarrow \frac{e^{2t}t^3}{3!}+5\frac{e^{2t}t^4}{4!}+8\frac{e^{2t}t^5}{5!}$$

(13) $\frac{s^2+1}{(s-1)^5}$

Sol:

$$\frac{(s^2-1+1)^2+1}{(s-1)^5}$$

$$\Rightarrow \frac{(s-1)^2+1+2(s-1)+1}{(s-1)^5} \Rightarrow \frac{(s-1)^2}{(s-1)^5}+\frac{2(s-1)}{(s-1)^5}+\frac{2}{(s-1)^5}$$

$$\Rightarrow \frac{1}{(s-1)^3}+\frac{2}{(s-1)^4}+\frac{2}{(s-1)^5}$$

$$\Rightarrow \frac{e^t t^2}{2!}+\frac{2t^3e^t}{3!}+2\frac{e^t t^4}{4!}$$

$\Rightarrow L^{-1}\left\{-\frac{dF(s)}{ds}\right\} = -2\cosh at + 2$

Divide by t

$f(t) = \frac{2 - 2\cosh at}{t}$

(2) ~~Same log of~~ $\log\left(\frac{s^2+1}{s(s+1)}\right)$

Sol: $\log(s^2+1) - \log s(s+1)$

$\Rightarrow \log(s^2+1) - \{\log s + \log(s+1)\}$

$\Rightarrow \log(s^2+1) - \log s - \log(s+1)$

Diff. w.r.t s.

$\frac{dF(s)}{ds} = \frac{1}{(s^2+1)} \times 2s - \frac{1}{s} - \frac{1}{(s+1)}$

Multiply by $-ve$ on b.s.

$-\frac{dF(s)}{ds} = \frac{-2s}{(s^2+1)} + \frac{1}{s} + \frac{1}{(s+1)}$

Apply L^{-1} on b.s.

$L^{-1}\left\{-\frac{dF(s)}{ds}\right\} = -2\cos t + 1 + e^{-t}$

Divide by t

$f(t) = \frac{1 + e^{-t} - 2\cos t}{t}$

(3) $\log\frac{s}{(s-1)}$

Sol: $\log s - \log(s-1)$

Diff w.r.t s

$\Rightarrow \frac{dF(s)}{ds} = \frac{1}{s} - \frac{1}{(s-1)}$

Multiply By $-ve$ on b.s

$-\frac{dF(s)}{ds} = -\frac{1}{s} + \frac{1}{(s-1)}$

$L^{-1}\left\{-\frac{dF(s)}{ds}\right\} = -1 + e^{t}$

$f(t) = \frac{e^t - 1}{t}$

DIVISION BY POWERS OF 's'

If $L^{-1}\{F(s)\} = f(t)$, then $L^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t f(t)\,dt$

NOTE: ① $L^{-1}\left\{\frac{F(s)}{s^3}\right\} = \int_0^t \int_0^t \int_0^t f(t)\,dt$

② $L^{-1}\left\{\frac{F(s)}{s^2}\right\} = \int_0^t \int_0^t f(t)\,dt$

Q Find $f(t)$ for the following ——

(Ex 1) ① $\frac{1}{s(s^2+a^2)}$

Sol' Consider $F(s) = \frac{1}{s(s^2+a^2)}$

$L^{-1}\{F(s)\} = \frac{1}{a}\sin at$

t

Now, $L^{-1}\left\{\frac{1}{s(s^2+a^2)}\right\} = \int_0^t \frac{1}{a}\sin at\,dt$

$= \frac{1}{a}\left[\frac{-\cos at}{a}\right]_0^t$

$= \frac{1}{a^2}\left[-\cos at + \cos 0\right] \Rightarrow \frac{1}{a^2}\left[1-\cos at\right]$

(Ex 1) ② $\frac{1}{s(s+a)^3}$

Sol' Consider, $F(s) = \frac{1}{(s+a)^3}$

Change $(s+a) \longrightarrow s$

$L^{-1}\{F(s)\} = e^{-at} L^{-1}\left\{\frac{1}{s^3}\right\}$

(7) $\frac{se^{-s/2}+\pi e^{-s}}{s^2+\pi^2}$

Sol: $\frac{se^{-s/2}}{s^2+\pi^2} + \frac{\pi e^{-s}}{s^2+\pi^2}$

$F(s_2) = \frac{\pi}{s^2+\pi^2}$

$\Rightarrow F(s_1) = \frac{s}{s^2+\pi^2}$

$L^{-1}\{F(s_1)\} = \cos\pi t = f(t)$

$f(t-1/2) = \cos\pi(t-1/2)$

$= \cos(\pi t - \pi/2)$

$= \cos(-(\pi/2 - \pi t))$

$= \cos(\pi/2 - \pi t)$

$f(t-1/2) = \sin\pi t$

$L^{-1}\{F(s_2)\} = \sin\pi t = f(t)$

$f(t-1) = \sin\pi(t-1)$

$= \sin(\pi t - \pi)$

$= \sin(-(\pi - \pi t))$

$= -\sin(\pi - \pi t)$

$f(t-1) = -\sin\pi t$

$\Rightarrow L^{-1}\left\{\frac{se^{-s/2}+\pi e^{-s}}{s^2+\pi^2}\right\} = \left\{\sin\pi t\, u(t-1/2)\right\} + \left\{-\sin\pi t\, u(t-1)\right.$

(8) $L^{-1}\left\{e^{-2\pi s}\frac{s^2-a^2}{(s^2+a^2)^2}\right\}$

Sol: Consider,

$F(s) = \frac{s^2-a^2}{(s^2+a^2)^2}$

$L^{-1}\{F(s)\} = t\cos at = f(t)$

$f(t-2\pi) = (t-2\pi)\cos a(t-2\pi)$

$= (t-2\pi)(\cos(at-2a\pi))$

$L^{-1}\left\{e^{-2\pi s}\frac{s^2-a^2}{(s^2+a^2)^2}\right\} = \left\{(t-2\pi)\cos(at-2a\pi)\, u(t-2\pi)\right\}$

Note:

$$\left[\because \int e^{ax}\cos bx\,dx = e^{ax}\left[\frac{a\cos bx + b\sin bx}{a^2+b^2}\right]\right]$$

Comparing with $\int e^{ax}\cos bx\,dx$

$a=2,\ b=3,\ x=v$

$$\therefore e^{-2t}\left[e^{2v}\left[\frac{2\cos 3v + 3\sin 3v}{2^2+3^2}\right]\right]_0^t$$

$$\Rightarrow \frac{e^{-2t}}{13}\left[e^{2t}(2\cos 3t + 3\sin 3t) - (1(2+0))\right]$$

$$\Rightarrow \frac{1}{13}\left[2\cos 3t + 3\sin 3t - 2e^{-2t}\right]$$

(3) $\frac{1}{s^2(s^2+1)}$

Sol: $\left(\frac{1}{s^2}\right)\left(\frac{1}{s^2+1}\right)$

$F(s)$ $\quad$ $G(s)$

$f(t) = t$ $\quad$ $g(t) = \sin t$

$f(v) = v$ $\quad$ $g(v) = \sin v$

$f(t-v) = (t-v)$

D	I
$(t-v)$	$\sin v$
-1	$-\cos v$
0	$\sin v$

(+, −)

Now,

$$f(t) * g(t) = \int_0^t f(t-v)\,g(v)\,dv = \int_0^t (t-v)\sin v\,dv$$

$$= \left[-(t-v)\cos v + \sin v\right]_0^t$$

$$= \left[0 + \sin t - \{-t(1) + 0\}\right]$$

$$= -\sin t + t$$

$$= \int_0^t \frac{1}{a} \sin(at-au)\cos au \, du$$

$$= \frac{1}{2a}\int_0^t \left[\sin at + \sin(at-2au)\right] du \quad \left[\sin A\cos B = \frac{1}{2}\left[\sin(A+B)+\sin(A-B)\right]\right]$$

$$= \frac{1}{2a}\left[\sin at(u) + \left(\frac{-\cos(at-2au)}{-2a}\right)\right]_0^t$$

$$= \frac{1}{2a}\left[\sin at(u) + \frac{\cos(at-2au)}{2a}\right]_0^t$$

$$= \frac{1}{2a}\left[t\sin at + \frac{\cos(at-2at)}{2a} - \left\{0 + \frac{\cos at}{2a}\right\}\right]$$

$$= \frac{1}{2a}\left[t\sin at + \frac{\cos at}{2a} \times 2\right] = \frac{1}{2a}\left[t\sin at + \frac{\cos at}{a}\right]$$

(6) $\frac{1}{(s^2+a^2)^2} = \frac{1}{(s^2+a^2)} * \frac{1}{(s^2+a^2)}$

Sol: $F(s)$ $\quad$ $G(s)$

$f(t) = \frac{1}{a}\sin at \quad g(t) = \frac{1}{a}\sin at$

$f(t-u) = \frac{1}{a}\sin(at-au) \quad g(u) = \frac{1}{a}\sin au$

Consider,

$$f(t) * g(t) = \int_0^t f(t-u)\, g(u)\, du$$

$$= \int_0^t \frac{1}{a}\sin(at-au) \times \frac{1}{a}\sin au \, du = \frac{1}{a^2}\int_0^t \sin(at-au)\sin au \, du$$

$$= \frac{-1}{2a^2}\int \left[\cos at - \cos(at-2au)\right] du$$

$$= \frac{-1}{2a^2}\left[\cos at(u) - \left(\frac{\sin(at-2au)}{-2a}\right)\right]_0^t$$

$\times (s-3)(s-1)(s-2)$

$\Rightarrow 1 = A(s-3)(s-1) + B(s-3)(s-2) + C(s-1)(s-2)$ ——— (2)

Considers,

$\Rightarrow \frac{1}{(s-3)(s-1)(s-2)} = \frac{A}{(s-2)}$ $\Rightarrow \frac{1}{(2-3)(2-1)} = A$

$s-2$

$\boxed{A = -1}$

$\Rightarrow \frac{1}{(s-3)(s-1)(s-2)} = \frac{B}{(s-1)}$ $\Rightarrow \frac{1}{(1-3)(1-2)} = B$

$s-1$

$\boxed{B = \frac{1}{2}}$

$\Rightarrow \frac{1}{(s-3)(s-1)(s-2)} = \frac{C}{(s-3)}$ $\Rightarrow \frac{1}{(3-1)(3-2)} = C$

$s-3$

$\boxed{C = \frac{1}{2}}$

$\because \frac{1}{(s-3)(s-1)(s-2)} = \frac{-1}{(s-2)} + \frac{1/2}{(s-1)} + \frac{1/2}{(s-3)}$

$\boxed{L^{-1}\{f(t)\} = -e^{2t} + \frac{1}{2}e^{t} + \frac{1}{2}e^{3t}}$ ——— (*)

Considers, $L^{-1}\left\{\frac{1}{(s-1)}\right\} = e^{t}$

$\Rightarrow y(s) = 3 + 2t + e^{2t} - 4e^{t} - e^{2t} + \frac{1}{2}e^{t} + \frac{1}{2}e^{3t} + e^{t}$

Consider,

$$\frac{4}{s^2(s-2)(s-1)} = \frac{A}{s^2} + \frac{B}{s^2} + \frac{C}{(s-2)} + \frac{D}{(s-1)} \longrightarrow (1)$$

$$\Rightarrow 4 = As(s-2)(s-1) + B(s-2)(s-1) + Cs^2(s-1) + Ds^2(s-2) \longrightarrow (2)$$

Consider,

$$\Rightarrow \frac{4}{s^2(s-2)(s-1)}\Big|_{s=2} = \frac{C}{(s-2)} \Rightarrow \frac{4}{2^2(2-1)} = C$$

$$C = \frac{4}{4 \times 1} = 1$$

$$\boxed{C = 1}$$

$$\Rightarrow \frac{4}{s^2(s-2)(s-1)}\Big|_{s=0} = \frac{B}{s^2} \Rightarrow \frac{4}{(0-2)(0-1)} = B$$

$$\boxed{B = 2}$$

$$\Rightarrow \frac{4}{s^2(s-2)(s-1)}\Big|_{s=1} = \frac{D}{(s-1)} \Rightarrow \frac{4}{1^2(1-2)} = D$$

$$\boxed{D = -4}$$

From eqn (2), Comparing coefficient of s^3 ⟶

$$\Rightarrow 0 = A + C + D$$

$$0 = A + 1 - 4$$

$$\boxed{A = 3}$$

$$\therefore \frac{4}{s^2(s-2)(s-1)} = \frac{3}{s} + \frac{2}{s^2} + \frac{1}{(s-2)} - \frac{4}{s-1}$$

$$\boxed{L^{-1}\{f(t)\} = 3 + 2t + e^{2t} - 4e^{t}} \longrightarrow (*)$$

Consider,

$$(2) \Rightarrow \frac{1}{(s-3)(s-1)(s-2)} = \frac{A}{(s-2)} + \frac{B}{(s-1)} + \frac{C}{(s-3)} \longrightarrow (1)$$

(2) $L\{f'\} = sF(s) - f(0)$

Also, $L\{f(t)\} = F(s)$

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Q Solve initial value problem——

(1) $\frac{d^2y}{dt^2} + a^2y = \cos at$ Given: $y=0$, $\frac{dy}{dt} = 0$ at $t=0$

Sol: $L\left\{\frac{d^2y}{dt^2}\right\} + a^2L\{y\} = L\{\cos at\}$

$s^2y(s) - sy(0) - y'(0) + a^2[y(s)] = \frac{s}{s^2+a^2}$

$[s^2+a^2]y(s) = \frac{s}{s^2+a^2}$

$y(s) = \frac{s}{(s^2+a^2)^2}$

$y(t)$
$y(0) = 0$
$y'(0) = 0$

$L^{-1}\{y(s)\} = L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\}$

$\boxed{y(t) = \frac{1}{2a} t \sin at}$

(2) $y'' - 3y' + 2y = 4t + e^{3t}$, $y(0) = 1$, $y'(0) = 1$

Sol: $L\{y''\} - 3L\{y'\} + 2L\{y\} = L\{4t + e^{3t}\}$

$\Rightarrow s^2y(s) - sy(0) - y'(0) - 3[y(s) - y(0)] + 2[y(s)] = \frac{4}{s^2} + \frac{1}{(s-3)}$

$\Rightarrow s^2y(s) - \underbrace{sy(0)}_{1} - \underbrace{y'(0)}_{1} - 3y(s) + \underbrace{3y(0)}_{1} + 2y(s) = \frac{4}{s^2} + \frac{1}{(s-3)}$

$\Rightarrow [s^2 - 3s + 2]y(s) = \frac{4}{s^2} + \frac{1}{(s-3)} + s - 2$

$\Rightarrow [(s-2)(s-1)]y(s) = \frac{4}{s^2} + \frac{1}{(s-3)} + s - 2$

$\Rightarrow y(s) = \underbrace{\frac{4}{s^2(s-2)(s-1)}}_{(1)} + \underbrace{\frac{1}{(s-3)(s-1)(s-2)}}_{(2)} + \frac{(s-2)}{(s-2)(s-1)}$

$$= \frac{-1}{2a^2}\left[\cos at(0) + \frac{\sin(at-2a0)}{2a}\right]_0^t$$

$$= \frac{-1}{2a^2}\left[\left\{t\cos at + \frac{\sin(at-2at)}{2a}\right\} - \left\{0 + \frac{\sin at}{2a}\right\}\right]$$

$$= \frac{-1}{2a^2}\left[t\cos at - \frac{\sin at}{2a} - \frac{\sin at}{2a}\right]$$

$$= \frac{-1}{2a^2}\left[t\cos at - \frac{\sin at}{a}\right]$$

⇒ APPLICATION OF LAPLACE TRANSFORM

⇒ Solution of differential eqn using Laplace Transform

① $L^{-1}\left\{\frac{1s}{(s^2+a^2)^2}\right\} = \frac{1}{2a} t\sin at$

② $L^{-1}\left\{\frac{s^2-a^2}{(s^2+a^2)^2}\right\} = t\cos at$

③ $L^{-1}\left\{\frac{d^2y}{dt^2}\right\} = L\{y''\} = s^2y(s) - sy(0) - y'(0)$

④ $L^{-1}\left[\frac{d^2x}{dt^2}\right] = L[x''] = s^2x(s) - sx(0) - x'(0)$

⑤ $L\{f''(t)\} = s^2F(s) - sF(0) - F'(0)$

⑥ $L\left\{\frac{dy}{dt}\right\} = L\{y'\} = sy(s) - y(0)$

Also, $L\{y\} = y(s)$

⑦ $L\left\{\frac{dx}{dt}\right\} = L\{x'\} = sx(s) - x(0)$

Also, $L\{x\} = x(s)$

④ $\frac{1}{(s+a)(s+b)} = \frac{1}{(s+a)} \cdot \frac{1}{(s+b)}$

Sol:

$F(s)$ $\quad$ $G(s)$

$f(t) = e^{-at}$ $\quad$ $g(t) = e^{-bt}$

$f(t-b) = e^{-a(t-v)}$ $\quad$ $g(v) = e^{-bv}$

$f(t-v) = e^{-at}e^{av}$

$$f(t) * g(t) = \int_0^t f(t-v)\, g(v)\, dv$$

$$= \int_0^t e^{-at} e^{av} e^{-bv}\, dv$$

$$= e^{-at} \int_0^t e^{(a-b)v}\, dv \quad = e^{-at} \times \left[\frac{e^{(a-b)v}}{(a-b)}\right]_0^t$$

$$= \frac{e^{-at}}{(a-b)}\left[e^{(a-b)t} - 1\right]$$

$$= \frac{e^{-bt}}{(a-b)} - \frac{e^{-at}}{(a-b)}$$

⑤ $\frac{s}{(s^2+a^2)^2} = \frac{s}{(s^2+a^2)} \times \frac{1}{(s^2+a^2)}$

$F(s)$ $\quad$ $G(s)$

$f(t) = \cos at$ $\quad$ $g(t) = \frac{1}{a}\sin at$

$g(v) = \cos av$ $\quad$ $g(t-v) = \frac{1}{a}\sin a(t-v)$

$= \frac{1}{a}\sin(at - av)$

$$f(t) * g(t) = \int_0^t g(t-v) f(v)\, dv$$

$$= \left[-(t-v)\frac{e^{-2v}}{2} + \frac{e^{-2v}}{4} \right]_0^t$$

$$= \left\{ \left[-(t-t)\frac{e^{-2t}}{2} + \frac{e^{-2t}}{4} \right] - \left[-\frac{t}{2} + \frac{1}{4} \right] \right\}$$

D	I
$(t-v)$	e^{-2v}
-1	$\frac{e^{-2v}}{2}$
0	$\frac{e^{-2v}}{4}$

$$\Rightarrow \frac{e^{-2t}}{4} + \frac{t}{2} - \frac{1}{4}$$

(2) $\frac{s}{(s+2)(s^2+9)}$

Sol: $\frac{1}{(s+2)} \cdot \frac{s}{(s^2+9)}$

$F(s)$ $\quad$ $G(s)$

$f(t) = e^{-2t}$ $\quad$ $g(t) = \cos 3t$

$g(v) = \cos 3v$

$f(t-v) = e^{-2(t-v)}$

$f(t-v) = e^{-2t}e^{2v}$

Now,

$$f(t) * g(t) = \int_0^t f(t-v)\, g(v)\, dv$$

$$= \int_0^t e^{-2t} e^{2v} \cos 3v \, dv$$

$$= e^{-2t} \int_0^t e^{2v} \cos 3v \, dv$$

WORKING RULE

1] Split the given function into two function (such that L^{-1} should exist)

2] Take one function as $F(s)$ and $G(s)$

3] Find $L^{-1}\{F(s)\}$ and $L^{-1}\{G(s)\}$ to get $f(t)$ and $g(t)$

4] Replace one function by v and another function by $(t-v)$

5] Integrate w.r.t v by taking limits $0 \longrightarrow t$

6] We get $L^{-1}\{F(s)G(s)\} = f(t) * g(t)$

CONVOLUTION :→ Used in LTI → Linear Time Invariant

$$f(t) * g(t) = \int_0^t f(v)\, g(t-v)\, dv$$

- Convolution is a mathematical operation on two functions say f and g producing a third function that is viewed as modified version of one of the original function.

① $e^t = e^{t-v}$

② $t \sin(t-v)$

$$f(t) * g(t) = \int_0^t f(v)\, g(t-v)\, dv$$

Note: ① Operation * is Commutative

② If $L^{-1}\{f(s)\} = f(t)$ and $L^{-1}\{g(s)\} = g(t)$, then

$$L^{-1}\{f(s)\, G(s)\} = \int_0^t f(v)\, g(t-v)\, dv = F(t) * g(t)$$

Q Using Convolution theorem Obtain the inverse Laplace transform of the following function.

① $\frac{1}{s(s+2)} = \frac{1}{s^2} \times \frac{1}{(s+2)}$

Sol: Consider, $F(s) = \frac{1}{s^2}$ $\quad G(s) = \frac{1}{(s+2)}$

$L^{-1}\{F(s)\} = t = f(t)$ $\quad L^{-1}\{G(s)\} = e^{-2t} = g(t)$

$f(t-v) = (t-v)$ $\quad g(v) = e^{-2v}$

Now,

$$f(t) * g(t) = \int_0^t f(t-v)\, g(v)\, dv = \int_0^t (t-v)\, e^{-2v}\, dv$$

(4) $\frac{e^{-s}}{(s+1)(s-2)}$

Sol: $F(s) = \frac{1}{(s+1)(s-2)}$

Numerator $1 = \frac{1}{3}[(s+1)-(s-2)]$

Now, $\frac{1}{3}\left[\frac{(s+1)-(s-2)}{(s+1)(s-2)}\right]$

$\Rightarrow \frac{1}{3}\left[\frac{1}{(s-2)} - \frac{1}{(s+1)}\right]$

$L^{-1}\{F(s)\} = \frac{1}{3}\left[e^{2t} - e^{-t}\right] = f(t)$

$f(t-1) = \frac{1}{3}\left(e^{2(t-1)} - e^{-(t-1)}\right)$

$L^{-1}\left\{\frac{e^{-s}}{(s+1)(s-2)}\right\} = \left[\frac{1}{3}\left[e^{2(t-1)} - e^{1-t}\right] U(t-1)\right]$

(5) $\frac{e^{-\pi s}\, s}{(s^2+a^2)^2}$

Sol: $F(s) = \frac{s}{(s^2+a^2)^2}$

$L^{-1}\{F(s)\} = \frac{1}{2a} t \sin at = f(t)$

$f(t-\pi) = \frac{1}{2a}(t-\pi)\sin(t-\pi)a$

$L^{-1}\left\{\frac{e^{-\pi s}\, s}{(s^2+a^2)^2}\right\} = \left\{\frac{1}{2a}(t-\pi)\sin(t-\pi)a\; U(t-\pi)\right\}$

(6) $\frac{1-e^{-\pi s}}{(s^2+4)}$

Sol: $F(s) = \frac{1}{(s^2+4)} - \frac{e^{-\pi s}}{(s^2+4)}$

$L^{-1}\{F(s)\} = \frac{1}{2}\sin 2t - \sin 2t$

$U(t-\pi)$
$\sin(2(t-\pi))$
$\sin(2t-2\pi)$
$\sin(-(2\pi-2t))$
$-\sin(2\pi-2t)$
$-\sin 2t$

$$= -4\left[\frac{4-3s^2}{(s^2+4)^3}\right] = \frac{12s^2-16}{(s^2+4)^3}$$

Q $L\{t^2\sin^2 3t\}$

Consider,

$f(t) = \sin^2 3t$

$\sin^2\theta = \frac{1-\cos 2\theta}{2}$

$\therefore \sin^2 3t = \frac{1-\cos 6t}{2}$

$$L\{f(t)\} = \frac{1}{2}L\{1-\cos 6t\} = \frac{1}{2}\left[\frac{1}{s} - \frac{s}{s^2+36}\right] = F(s)$$

$$= \frac{1}{2}\left[\frac{36}{s(s^2+36)}\right]$$

$\because$ WKT $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$

$$\therefore L\{t^2\sin^2 3t\} = (-1)^2 \frac{d^2}{ds^2}\left[\frac{1}{2}\left(\frac{1}{s} - \frac{s}{s^2+36}\right)\right]$$

$$= \frac{d}{ds}\frac{d}{ds}\left[\frac{1}{s} - \frac{s}{s^2+36}\right]$$

$$= \frac{1}{2}\frac{d}{ds}\left[-\frac{1}{s^2} - \left\{\frac{(s^2+36) - s(2s)}{(s^2+36)^2}\right\}\right]$$

$$= \frac{1}{2}\frac{d}{ds}\left[-\frac{1}{s^2} - \left\{\frac{s^2+36-2s^2}{(s^2+36)^2}\right\}\right]$$

$$= \frac{1}{2}\frac{d}{ds}\left[-\frac{1}{s^2} - \left\{\frac{-s^2+36}{(s^2+36)^2}\right\}\right]$$

$$= \frac{1}{2}\frac{d}{ds}\left[-\frac{1}{s^2} + \frac{(s^2-36)}{(s^2+36)^2}\right]$$

$$= \frac{1}{2}\left[\frac{2}{s^3} + \left[\frac{(s^2+36)^2 \times 2s - 2(s^2-36)(s^2+36)\times 2s}{(s^2+36)^4}\right]\right]$$

$$= \frac{1}{2}\left[\frac{2}{s^3} + \left\{(s^2+36)\left[\frac{2s(s^2+36) - 4s(s^2-36)}{(s^2+36)^3}\right]\right\}\right]$$

④ $\log\left(\frac{s^2+a^2}{(s-a)^2}\right)$

Sol: $\log(s^2+a^2) - \log(s-a)^2$

$\Rightarrow \log(s^2+a^2) - 2\log(s-a)$

Diff w.r.t s

$$\frac{dF(s)}{ds} = \frac{1}{(s^2+a^2)} \times 2s - \frac{2(1)}{(s-a)}$$

$$\frac{dF(s)}{ds} = \frac{2s}{(s^2+a^2)} - \frac{2}{(s-a)}$$

Multiply by -ve on b.s.

$$\frac{-dF(s)}{ds} = \frac{-2s}{(s^2+a^2)} + \frac{2}{(s-a)}$$

Apply L^{-1} on b.s.

$$L^{-1}\left\{\frac{-dF(s)}{ds}\right\} = -2L^{-1}\left\{\frac{s}{s^2+a^2}\right\} + 2L^{-1}\left\{\frac{1}{s-a}\right\}$$

$$= -2\cos at + 2e^{at}$$

Divide by t

$$f(t) = \frac{2e^{at} - 2\cos at}{t}$$

⑤ $\tan^{-1}\left(\frac{a}{s}\right) = F(s)$

Sol: $$\frac{dF(s)}{ds} = \frac{1}{1+\left(\frac{a}{s}\right)^2} \cdot \left(\frac{-a}{s^2}\right)$$

$$= \frac{1}{\frac{s^2+a^2}{s^2}} \times \frac{-a}{s^2}$$

$$\frac{dF(s)}{ds} = \frac{-a}{s^2+a^2}.$$

Multiply by -ve on bs

$$\frac{-dF(s)}{ds} = \frac{a}{s^2+a^2}$$

INVERSE TRANSFORM OF TRIGNOMETRIC (tan and cot) AND LOG FUNCTIONS

If $L\{f(t)\} = F(s)$ then $L\{t(f(t))\} = -\frac{d}{ds}(F(s))$

WORKING RULE

Step①: Diff. w.r.t s i.e, $\frac{dF(s)}{ds}$

Step②: Multiply by -ve sign i.e, $-\frac{dF(s)}{ds}$

Step③: Apply L^{-1} to both sides i.e, $L^{-1}\left\{-\frac{dF(s)}{ds}\right\} = tf(t)$

Step④: Divide by t to get $f(t)$

Q Find the $f(t)$ for the following —

① $\log\left(1-\frac{a^2}{s^2}\right)$

Sol: $\log\left(\frac{s^2-a^2}{s^2}\right)$

$\Rightarrow \log(s^2-a^2) - \log s^2$

$\Rightarrow \log(s^2-a^2) - 2\log s$

Diff w.r.t s

$\Rightarrow \frac{dF(s)}{ds} = \frac{1}{(s^2-a^2)} \times 2s - \frac{2\times 1}{s}$

Multiply by -ve on B.s.

$\Rightarrow -\frac{dF(s)}{ds} = \frac{-2s}{(s^2-a^2)} + \frac{2}{s}$

Apply L^{-1} on B.s

$\Rightarrow L^{-1}\left\{-\frac{dF(s)}{ds}\right\} = -2L^{-1}\left\{\frac{s}{s^2-a^2}\right\} + L^{-1}\left\{\frac{2}{s}\right\}$

(14) $\frac{3s-2}{(2s-1)^7} = \frac{3s-2}{[2(s-\frac{1}{2})]^7}$

Sol: $\Rightarrow \frac{1}{2^7} \frac{(3s-2)}{(s-\frac{1}{2})^7}$

$\Rightarrow \frac{1}{2^7}\left[\frac{3s}{(s-\frac{1}{2})^7} - \frac{2}{(s-\frac{1}{2})^7}\right]$

$\Rightarrow \frac{1}{2^7}\left[3e^{\frac{1}{2}t}\frac{t^6}{6!}\right] \Rightarrow \frac{1}{2^7}\left[3(s-\frac{1}{2}+\frac{1}{2}) - 2\right]$

$\Rightarrow \frac{1}{2^7}\left[\frac{3(s-\frac{1}{2})}{(s-\frac{1}{2})^7} - \frac{\frac{1}{2}}{(s-\frac{1}{2})^7}\right]$

$\Rightarrow \frac{1}{2^7}\left[\frac{3}{(s-\frac{1}{2})^6} - \frac{\frac{1}{2}}{(s-\frac{1}{2})^7}\right] \Rightarrow \frac{1}{2^7}\left[3e^{\frac{1}{2}t}\frac{t^5}{5!} - \frac{1}{2}e^{\frac{1}{2}t}\frac{t^6}{6!}\right]$

(15) $\frac{s^2}{(s+a)^3}$

Sol: $\frac{(s+a-a)^2}{(s+a)^3} = \frac{(s+a)^2+a^2-2a(s+a)}{(s+a)^3}$

$= \frac{(s+a)^2}{(s+a)^3} - \frac{2a(s+a)}{(s+a)^3} + \frac{a^2}{(s+a)^3}$

$= \frac{1}{(s+a)} - \frac{2a}{(s+a)^2} + \frac{a^2}{(s+a)^3}$

$= e^{-at} - 2ae^{-at}\frac{t}{1!} + a^2e^{-at}\frac{t^2}{2!}$

$$= \frac{1}{2}\left[\frac{s}{s^2-a^2} + \frac{s}{s^2+a^2}\right]$$

$$f(t) = \frac{1}{2}\left[\cosh at + \cos at\right]$$

Note: We can evaluate $\frac{1}{(s^4-1)}$ and $\frac{s^3}{(s^4-1)}$ in the same manner.

(1) $\frac{s^2+6}{(s^2+4)(s^2+1)} = \frac{As+B}{s^2+4} + \frac{Cs+D}{s^2+1}$

Sol: Put $s^2 = x$

$$\frac{x+6}{(x+4)(x+1)} = \frac{A}{(x+4)} + \frac{B}{(x+1)} \quad \text{---(1)}$$

Compare, $\frac{x+6}{(x+4)(x+1)} = \frac{A}{(x+4)}$

$x = -4$

$\Rightarrow \frac{-4+6}{-4+1} = A \Rightarrow A = \frac{2}{-3}$

$\boxed{A = -\frac{2}{3}}$

Compare, $\frac{x+6}{(x+4)(x+1)} = \frac{B}{(x+1)}$

$x = -1$

$\Rightarrow \frac{-1+6}{-1+4} = B \Rightarrow \boxed{B = \frac{5}{3}}$

Substitute in (1) $\longrightarrow$

$$\frac{x+6}{(x+4)(x+1)} = \frac{-2/3}{(x+4)} + \frac{5/3}{(x+1)}$$

Substitute $x \longrightarrow s^2$

$$\frac{s^2+6}{(s^2+4)(s^2+1)} = \frac{-2/3}{(s^2+4)} + \frac{5/3}{(s^2+1)}$$

$$= \frac{-2}{3} \times \frac{1}{2}\sin 2t + \frac{5}{3}\sin t$$

$$f(t) = \frac{-1}{3}\sin 2t + \frac{5}{3}\sin t$$

$$= \frac{-1}{2a^2}\left[\frac{1}{a}\sin at - \frac{1}{a}\sin hat\right]$$

(9) $\frac{s^2}{s^4-a^4}$

Sol: $\frac{s}{(s^2)^2-(a^2)^2} = \frac{s}{(s^2-a^2)(s^2+a^2)}$

Numerator:- $\frac{1}{2a^2}\left[(s^2+a^2)-(s^2-a^2)\right]$

$\frac{s}{(s^2-a^2)(s^2+a^2)} = \frac{1}{2a^2}\left[\frac{(s^2+a^2)-(s^2-a^2)}{(s^2-a^2)(s^2+a^2)}\right]$

=

Numerator:- $\frac{1}{2s}\left[(s^2-a^2)+(s^2+a^2)\right]$

$\frac{s}{(s^2-a^2)(s^2+a^2)} = \frac{1}{2s}\left[\frac{(s^2-a^2)+(s^2+a^2)}{(s^2-a^2)(s^2+a^2)}\right]$

$= \frac{1}{2s}\left[\frac{1}{(s^2+a^2)} + \frac{1}{(s^2-a^2)}\right]$

(10) $\frac{s^3}{s^4-a^4}$

Sol: $\frac{s^3}{(s^2)^2-(a^2)^2} = \frac{s^3}{(s^2+a^2)(s^2-a^2)}$

$\rightarrow \frac{s^3}{(s^2+a^2)(s^2-a^2)} = \frac{s}{2}\left[\frac{(s^2+a^2)+(s^2-a^2)}{(s^2+a^2)(s^2-a^2)}\right]$

$= \frac{s}{2}\left[\frac{1}{s^2-a^2} + \frac{1}{s^2+a^2}\right]$

(7) $\frac{3s+5}{(s^2-5s+6)}$

Sol: s^2-5s+6

$\Rightarrow s^2-2s-3s+6$

$\Rightarrow s(s-2)-3(s-2)$

$\Rightarrow (s-3)(s-2)$

$-6s^2$
$-5s$
$-2s \quad -3s$

Now $\frac{3s+5}{(s-3)(s-2)} = \frac{A}{(s-3)} + \frac{B}{(s-2)}$

Consider,

$\frac{3s+5}{(s-3)(s-2)} = \frac{A}{(s-3)}$ $\Rightarrow \frac{3(3)+5}{3-2} = A$

$s=3$

$A = \frac{14}{1}$ $\boxed{A=14}$

$\frac{3s+5}{(s-3)(s-2)} = \frac{B}{(s-2)}$ $\Rightarrow \frac{3(2)+5}{2-3} = B$

$s=2$

$\boxed{B=-11}$

(7) $\Rightarrow \frac{3s+5}{(s-3)(s-2)} = \frac{14}{(s-3)} \quad \frac{-11}{(s-2)}$

$= 14e^{3t} - 11e^{2t}$

(8) $\frac{1}{s^4-a^4}$ (9) $\frac{s}{s^4-a^4}$

Sol: $\frac{1}{(s^2)^2-(a^2)^2}$

$\Rightarrow \frac{1}{(s^2-a^2)(s^2+a^2)}$

$\Rightarrow \frac{1}{(s+a)(s-a)(s^2+a^2)} \quad \frac{A}{(s+a)} + \frac{B}{(s-a)}$

Nr: $\frac{-1}{2a^2}\left[(s^2-a^2)-(s^2+a^2)\right]$

$\frac{1}{(s^2-a^2)(s^2+a^2)} = \frac{-1}{2a^2}\left[\frac{(s^2-a^2)-(s^2+a^2)}{(s^2-a^2)(s^2+a^2)}\right]$

$= \frac{-1}{2a^2}\left[\frac{1}{(s^2+a^2)} - \frac{1}{(s^2-a^2)}\right]$

Compare, $\left.\frac{s+5}{s^2(s-1)(s+2)}\right|_{s=1} = \frac{C}{(s-1)}$ $\Rightarrow \frac{1+5}{1^2(1+2)} = C$

$C = \frac{6}{3} = 2$

$\boxed{C=2}$

$\Rightarrow \left.\frac{s+5}{s^2(s-1)(s+2)}\right|_{s=-2} = \frac{D}{(s+2)}$ $\Rightarrow \frac{-2+5}{(-2)^2(-2-1)} = D$

$D = \frac{3}{4\times(-3)}$

$D = \frac{3}{-12} = \frac{-1}{4}$

$\boxed{D = -\frac{1}{4}}$

From (2)

Compare the coefficient of s^3

$0 = A + C + D$

$0 = A + 2 - \frac{1}{4}$

$A = \frac{1}{4} - 2 = \frac{-7}{4}$ $\rightarrow \boxed{A = -\frac{7}{4}}$

(1) $\Rightarrow \frac{s+5}{s^2(s-1)(s+2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{(s-1)} + \frac{D}{(s+2)}$

$= \frac{(-7/4)}{s} + \frac{(-5/2)}{s^2} + \frac{2}{(s-1)} + \frac{(-1/4)}{s+2}$

$= \frac{-7}{4} - \frac{5}{2}t + 2e^t - \frac{1}{4}e^{-2t}$

$= -\frac{7}{4} - \frac{5}{2}t + 2e^t - \frac{1}{4}e^{-2t}$

Find $f(t)$ for the following functions

(4) $\frac{3s+2}{(s-1)(s^2+1)}$

Sol: $\frac{3s+2}{(s-1)(s^2+1)} = \frac{A}{(s-1)} + \frac{Bs+C}{s^2+1}$ ——— (1)

$\times (s-1)(s^2+1)$

$\Rightarrow 3s+2 = A(s^2+1) + (Bs+C)(s-1)$ ——— (2)

Consider, $s-1=0$

$\frac{3s+2}{(s-1)(s^2+1)} = \frac{A}{(s-1)}$

$\Rightarrow \frac{3(1)+2}{(1^2+1)} = A$

$\Rightarrow \boxed{A = 5/2}$

From eqn (2), equate co-efficient of s^2

$0 = A+B$

$\boxed{B = -5/2}$

Put $s=0$ in eqn (2)

$0+2 = A+(-C)$

$C = A-2 \rightarrow \frac{5}{2}-2$

$\boxed{C = \frac{1}{2}}$

(1) $\Rightarrow \frac{3s+2}{(s-1)(s^2+1)} = \frac{5/2}{(s-1)} + \frac{-5/2 s + 1/2}{(s^2+1)}$

$= \frac{5/2}{(s-1)} - \frac{5}{2}\left[\frac{s}{s^2+1}\right] + \frac{1}{2}\left[\frac{1}{s^2+1}\right]$

$= \frac{5}{2}e^t - \frac{5}{2}\cos t + \frac{1}{2}\sin t$

Consider from ① —

$\Rightarrow \frac{(s+2)}{(s+1)^2(s-2)} = \frac{B}{(s+1)^2}$ $\rightarrow \frac{-1+2}{-1-2} = B \Rightarrow \boxed{B = -\frac{1}{3}}$

$s = -1$

$\Rightarrow \frac{s+2}{(s+1)^2(s-2)} = \frac{C}{(s-2)}$ $\rightarrow \frac{2+2}{(2+1)^2} = C \Rightarrow \boxed{C = \frac{4}{9}}$

$s = 2$

From ② — Compare coefficient of s^2

$0 = A + C$

$0 = A + \frac{4}{9}$

$\boxed{A = -\frac{4}{9}}$

① $\Rightarrow$

$$\frac{s+2}{(s+1)^2(s-2)} = \frac{(-4/9)}{(s+1)} + \frac{(-1/3)}{(s+1)^2} + \frac{(4/9)}{(s-2)}$$

$$= -\frac{4}{9}e^{-t} - \frac{1}{3}e^{-t}\frac{t}{\lfloor 2} + \frac{4}{9}e^{2t}$$

$$= -\frac{4}{9}e^{-t} - \frac{1}{3}e^{-t}xt + \frac{4}{9}e^{2t}$$

③ $L^{-1}\left\{\frac{1}{(s+1)^{10}}\right\}$

Change $(s+1)$ to s —

$= e^{-t}L^{-1}\left\{\frac{1}{s^{10}}\right\} = e^{-t}\frac{t^9}{\lfloor 10} = e^{-t}\frac{t^9}{10!}$ $\frac{e^t t^9}{9!}$

→ SECOND SHIFTING THEOREM

If $L^{-1}\{F(s)\} = f(t)$, then

$L^{-1}\{e^{-as}F(s)\} = \{f(t-a)\,U(t-a)\}$

Q Find the L^{-1} for the following

① $\frac{e^{-2s}\,s}{s^2+1}$

Sol: $L^{-1}\left\{\frac{e^{-2s}\,s}{s^2+1}\right\} = \{\cos(t-2)\,U(t-2)\}$

Considers, $F(s) = \frac{s}{s^2+1}$

$L^{-1}\{F(s)\} = \cos t \cdot f(t)$

$f(t-2) = \cos(t-2)$

② $\frac{e^{-s}}{s^3}$

Sol: Considers,

$F(s) = \frac{1}{s^3}$

$L^{-1}\{F(s)\} = \frac{t^2}{2!} = f(t)$

$f(t-1) = \frac{(t-1)^2}{2!}$

$\therefore L^{-1}\left\{\frac{e^{-s}}{s^3}\right\} = \left\{\frac{(t-1)^2}{2!}U(t-1)\right\}$

③ $\frac{e^{-3s}}{(s-1)^4}$

Considers,

$F(s) = \frac{1}{(s-1)^4}$

Change $(s-1)$ to s

$L^{-1}\{F(s)\} = \frac{e^t t^3}{3!} = f(t)$

$f(t-3) = \frac{e^{(t-3)}(t-3)^3}{3!}$

$L^{-1}\left\{\frac{e^{-3s}}{(s-1)^4}\right\} = \left\{\frac{e^{(t-3)}(t-3)^3}{3!}U(t\right.$

$L^{-1}\{F(s)\} = e^{-at}\frac{t^2}{2!}$

$\therefore L^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t e^{-at}\frac{t^2}{2!}\,dt$

$= -\frac{1}{2}\left[-t^2\frac{e^{-at}}{a} - 2t\frac{e^{-at}}{a^2} - 2\frac{e^{-at}}{a^3}\right]_0^t$

$= \frac{1}{2}\left[-t^2\frac{e^{-at}}{a} - 2t\frac{e^{-at}}{a^2} - 2\frac{e^{-at}}{a^3}\right.$

D	I
t^2	e^{-at}
$2t$	$\frac{e^{-at}}{-a}$
2	$\frac{e^{-at}}{a^2}$
0	$\frac{e^{-at}}{-a^3}$

(3) $\frac{s^2-a^2}{s(s^2+a^2)^2}$

Sol $F(s) = \frac{s^2-a^2}{(s^2+a^2)^2}$

$L^{-1}\{F(s)\} = t\cos at$

$\therefore L^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t t\cos at\,dt$

$= \left[t\frac{\sin at}{a} + \frac{\cos at}{a^2}\right]_0^t$

$= \left[t\frac{\sin at}{a} + \frac{\cos at}{a^2} - \frac{1}{a^2}\right]$

NOTE!

* $L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\} = \frac{1}{2a}t\sin at$

* $L^{-1}\left\{\frac{s^2-a^2}{(s^2+a^2)^2}\right\} = t\cos at$

D	I
t	$\cos at$
1	$\frac{\sin at}{a}$
0	$-\frac{\cos at}{a^2}$

(4) $\frac{s^2}{s(s^2+a^2)^2} \Rightarrow \frac{s^2}{s(s^2+a^2)^2}$

Sol $F(s) = \frac{s}{(s^2+a^2)^2}$ $\Rightarrow L^{-1}\{F(s)\} = L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\}$

$= \frac{1}{2a}t\sin at$

Apply L^{-1} on B.S.

$$L^{-1}\left\{\frac{-dF(s)}{ds}\right\} = L^{-1}\left\{\frac{a}{s^2+a^2}\right\}$$

$$L^{-1}\{F(s)\} = \sin at$$

Divide By t

$$f(t) = \frac{\sin at}{t}$$

(6) $\cot^{-1}(s+1) = F(s)$

Sol: $\frac{dF(s)}{ds} = -\frac{1}{1+(s+1)^2}$

Multiply by -ve on bs.

$$\frac{-dF(s)}{ds} = \frac{1}{1+(s+1)^2}$$

Apply L^{-1} on b.s

$$L^{-1}\left\{\frac{-dF(s)}{ds}\right\} = L^{-1}\left\{\frac{1}{1+(s+1)^2}\right\}$$

Change $(s+1)$ to s

$$L^{-1}\{F(s)\} = e^{-t}\sin t$$

Divide by t.

$$f(t) = \frac{e^{-t}\sin t}{t}$$

(7) $\cot^{-1}\left(\frac{s+3}{2}\right)$

Sol: $\frac{dF(s)}{ds} = -\frac{1}{1+\left(\frac{s+3}{2}\right)^2}\cdot\frac{1}{2}$

Multiply by -ve on b.s

$$\frac{-dF(s)}{ds} = \frac{1}{1+\left(\frac{s+3}{2}\right)^2}\cdot\frac{1}{2}$$

$$\frac{-dF(s)}{ds} = \frac{1}{4+(s+3)^2}\cdot\frac{1}{\cancel{4}\,2}$$

$$\frac{-dF(s)}{ds} = \frac{2}{4+(s+3)^2}$$

Apply L^{-1} on b.s.

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{2}{2^2+(s+3)^2}\right\}$$

Change $(s+3)$ to s

$$L^{-1}\{F(s)\} = e^{-3t}\sin 2t$$

Divide by t

$$f(t) = \frac{e^{-3t}\sin 2t}{t}$$

(8) $\log\left(\frac{s}{s^2-1}\right)$

Sol: $\log s - \log(s^2-1)$

Diff w.r.t s

$$\frac{dF(s)}{ds} = \frac{1}{s} - \frac{1}{(s^2-1)}\times 2s$$

Multiply by -ve on b.s.

$$\frac{-dF(s)}{ds} = \frac{-1}{s} + \frac{2s}{(s^2-1)}$$

Apply L^{-1} on b.s.

$$L^{-1}\{F(s)\} = -1 + 2\cosh t$$

Divide by t

$$f(t) = \frac{-1+2\cosh t}{t}$$

Now, Numerator —

$3s+7 = 3(s+1-1)+7$

$= 3(s-1)+10$

$\therefore \frac{3s+7}{s^2-2s-3} = \frac{3(s-1)+10}{(s-1)^2-(2)^2}$

$= \frac{3(s-1)}{(s-1)^2-(2)^2} + \frac{10}{(s-1)^2-(2)^2}$

Change $(s-1)$ to s —

$= 3e^t L^{-1}\left\{\frac{s}{s^2-2^2}\right\} + 10\, e^t L^{-1}\left\{\frac{1}{s^2-2^2}\right\}$

$= 3e^t \cosh 2t + \frac{10}{2} e^t \times \frac{1}{2} \sinh 2t$

$= 3e^t \cosh 2t + 5e^t \sinh 2t$

(6) $\frac{s-7}{s^2-4s-21}$

Sol: Completing the squares for the denominator —

$s^2-4s-21$

$\Rightarrow \left[\left(s-\frac{1}{2}(4)\right)^2 - 21 - (2)^2\right]$

$\Rightarrow \left[(s-2)^2 - 25\right] \Rightarrow (s-2)^2 - 5^2$

Now, numerator,

$(s-7) = (s-2+2)-7 = (s-2)-5$

$\frac{s-7}{s^2-4s-21} = \frac{(s-2)-5}{(s-2)^2-5^2} \Rightarrow \frac{(s-2)}{(s-2)^2-5^2} - \frac{5}{(s-2)^2-5^2}$

Change $(s-2)$ to s —

$\Rightarrow e^{2t} L^{-1}\left\{\frac{s}{s^2-5^2}\right\} - 5e^{2t} L^{-1}\left\{\frac{1}{s^2-5^2}\right\}$

$\Rightarrow e^{2t} \cosh(5t) - \frac{5}{5} e^{2t} \times \frac{1}{5} \sinh(5t)$

$$\rightarrow \frac{3s+5}{4s^2+12s+17} = \frac{3(s+\frac{3}{2})+\frac{1}{2}}{4\left[(s+\frac{3}{2})^2+(\sqrt{2})^2\right]}$$

$$= \frac{3(s+\frac{3}{2})}{4\left[(s+\frac{3}{2})^2+(\sqrt{2})^2\right]} + \frac{\frac{1}{2}}{4\left[(s+\frac{3}{2})^2+(\sqrt{2})^2\right]}$$

Change to $s+\frac{3}{2} \rightarrow s$

$$= \frac{3}{4}e^{-\frac{3}{2}t}L^{-1}\left\{\frac{s^2}{s^2+(\sqrt{2})^2}\right\} + \frac{1}{8}e^{-\frac{3}{2}t}L^{-1}\left\{\frac{1}{s^2+(\sqrt{2})^2}\right\}$$

$$= \frac{3}{4}e^{-\frac{3}{2}t}\cos\sqrt{2}t + \frac{1}{8}e^{-\frac{3}{2}t} \times \frac{1}{\sqrt{2}}\sin\sqrt{2}t$$

(5) $\frac{3s+7}{s^2-2s-3}$

Sol) Completing the square for the denominator —

s^2-2s-3

$$\Rightarrow \left[\left(s-\frac{1}{2}(2)\right)^2-3-(1)^2\right] = \left[(s-1)^2-4\right]$$

$$\Rightarrow (s-1)^2-(2)^2$$

Now, numerator —

$$3s+7 = 3(s+2-2)+7$$
$$= 3(s+2)+1$$

$$\therefore \frac{3s+7}{s^2-2s-3} = \frac{3(s+1)+1}{(s-1)}$$

$$\because \frac{7s+4}{4s^2+4s+9} = \frac{7\left(s+\frac{1}{2}\right)+\frac{1}{2}}{4\left[\left(s+\frac{1}{2}\right)^2+\left(2\sqrt{2}\right)^2\right]}$$

$$= \frac{7\left(s+\frac{1}{2}\right)}{4\left[\left(s+\frac{1}{2}\right)^2+\left(2\sqrt{2}\right)^2\right]} + \frac{\frac{1}{2}}{4\left[\left(s+\frac{1}{2}\right)^2+\left(2\sqrt{2}\right)^2\right]}$$

Change $\left(s+\frac{1}{2}\right)$ to s

$$\Rightarrow \frac{7}{4}e^{-\frac{1}{2}t}L^{-1}\left\{\frac{s}{s^2+\left(2\sqrt{2}\right)^2}\right\} + e^{\frac{1}{2}t}\times\frac{1}{8}L^{-1}\left\{\frac{1}{s^2+\left(2\sqrt{2}\right)^2}\right\}$$

$$\Rightarrow \frac{7}{4}e^{-\frac{1}{2}t}\cos 2\sqrt{2}t + \frac{1}{2}e^{-\frac{1}{2}t}\times\frac{1}{2\sqrt{2}}\sin 2\sqrt{2}t$$

$$\Rightarrow \frac{7}{4}e^{-\frac{1}{2}t}\cos\sqrt{2}t + \frac{1}{2\sqrt{2}}e^{-\frac{1}{2}t}\sin\sqrt{2}t$$

(4) $\frac{3s+5}{4s^2+12s+17}$

Sol: Completing the square for the denominator

$$4s^2+12s+17$$

$$\Rightarrow 4\left[s^2+\frac{12}{4}s+\frac{17}{4}\right] \Rightarrow 4\left[s^2+3s+\frac{17}{4}\right]$$

$$\Rightarrow 4\left[\left(s+\frac{1}{2}(3)\right)^2+\frac{17}{4}-\frac{9}{4}\right] \Rightarrow 4\left[\left(s+\frac{3}{2}\right)^2+\left(\frac{8}{4}\right)\right]$$

$$\Rightarrow 4\left[\left(s+\frac{3}{2}\right)^2+\left(\sqrt{2}\right)^2\right]$$

Num: $3s+5 = 3\left(s+\frac{3}{2}-\frac{3}{2}\right)+5 \Rightarrow 3\left(s+\frac{1}{2}\right)+\frac{7}{2}$

$$= 3\left(s+\frac{1}{2}\right)+\frac{1}{2}$$

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Q s^2-4s+8

⇒ Comparing with ax^2+bx+c.

$\therefore \left[\left(s-\frac{1}{2}(4)\right)^2+8-(2)^2\right]$

→ $(s-2)^2+4$

Q $s^2+6s+18$

⇒ $\therefore \left[\left(s+\frac{1}{2}(6)\right)^2+18-(3)^2\right]$

→ $\left[(s+3)^2+9\right] = (s+3)^2+9$

Q s^2-2s-3

⇒ $\therefore \left[\left(s-\frac{1}{2}(2)\right)^2-3-(1)^2\right]$

→ $(s-1)^2-4$

Q Obtain a function of t for the following ——

(1) $\frac{s-2}{s^2-4s+8}$

⇒ Completing the square for the denominator.

s^2-4s+8

⇒ $\left[\left(s-\frac{1}{2}(4)\right)^2+8-(2)^2\right] \Rightarrow (s-2)^2+4$

Now, $\frac{s-2}{(s-2)^2+4} \Rightarrow \frac{s-2}{(s-2)^2+2^2}$

$\Rightarrow e^{2t}\cos 2t$

∴ Change $s-2$ to s

$= e^{2t}\left\{L^{-1}\left[\frac{s}{s^2+2^2}\right]\right\}$

$= e^{2t}\cos 2t$

Q Find f(t) for the following function —

① $\frac{8s}{(4s^2+1)^2}$

Sol: $\frac{8s}{[4(s^2+1/4)]^2} \Rightarrow \frac{8s}{16(s^2+1/4)^2} \Rightarrow \frac{s}{2(s^2+1/4)}$

$\Rightarrow \frac{1}{2} t \sin\left(\frac{1}{2}t\right)$

$\Rightarrow t\sin\left(\frac{t}{2}\right)$

② $\frac{9s^2-1}{(9s^2+1)^2}$

Sol: $\frac{9(s^2-1/9)}{[9(s^2+1/9)]^2} \Rightarrow \frac{9(s^2-1/9)}{81(s^2+1/9)^2} \Rightarrow \frac{1}{9}\left[\frac{s^2-(1/3)^2}{(s^2+(1/3)^2)^2}\right]$

$\Rightarrow \frac{1}{9} t\cos\frac{1}{3}t \Rightarrow \frac{1}{9}t\cos\frac{1}{3}t$

NOTE! Completing the squares —

$\Rightarrow ax^2+bx+c$

$a\left[x^2+\frac{b}{a}x+\frac{c}{a}\right] \Rightarrow a\left[\left(x+\frac{1}{2}\frac{b}{a}\right)^2+\frac{c}{a}-\left(\frac{b}{2a}\right)^2\right]$

(OR)

$\Rightarrow ax^2+bx+c$

$a\left[x^2+\frac{b}{a}x+\frac{c}{a}\right]$

$= a\left[\left(x+\frac{1}{2}\text{Coefficient of } x\right)^2+\text{Constant}-\left(\frac{1}{2}\text{coefficient } x\right)^2\right]$

$\Rightarrow a\left[\left(x+\frac{1}{2}\frac{b}{a}\right)^2+\frac{c}{a}-\left(\frac{b}{2a}\right)^2\right]$

⑦ $\frac{s+1}{s^{4/3}}$

$\Rightarrow \frac{s}{s^{4/3}} + \frac{1}{s^{4/3}}$

$\Rightarrow \frac{1}{s^{1/3}} + \frac{1}{s^{4/3}}$

$\Rightarrow \frac{t^{1/3-1}}{\Gamma(1/3)} + \frac{t^{4/3-1}}{\Gamma(4/3)}$

$\Rightarrow \frac{t^{-2/3}}{\Gamma(1/3)} + \frac{t^{1/3}}{\Gamma(1/3+1)} \Rightarrow \frac{t^{-2/3}}{\Gamma(1/3)} + \frac{t^{1/3}}{\frac{1}{3}\Gamma(1/3)}$

Q Find the f(t) for the following —

① $\left(\frac{\sqrt{s}-1}{s}\right)^2$

Sol $\frac{(\sqrt{s}-1)^2}{s^2} = \frac{s+1-2\sqrt{s}}{s^2}$

$\Rightarrow \frac{s}{s^2} + \frac{1}{s^2} - \frac{2\sqrt{s}}{s^2}$

$\Rightarrow \frac{1}{s} + \frac{1}{s^2} + \frac{-2}{s^{3/2}} = 1 + \frac{t^1}{\Gamma(2)} - \frac{2t^{1/2}}{\Gamma(\frac{3}{2}+1)}$

$\Rightarrow 1 + t - 4\frac{\sqrt{t}}{\sqrt{\pi}}$

② $\frac{(s+1)(2-\sqrt{s})}{s^{5/2}}$

Sol $\frac{2s - s^{3/2} + 2 - s^{1/2}}{s^{5/2}} \Rightarrow \frac{2s}{s^{5/2}} - \frac{s^{3/2}}{s^{5/2}} + \frac{2}{s^{5/2}} - \frac{s^{1/2}}{s^{5/2}}$

$\Rightarrow \frac{2}{s^{3/2}} - \frac{1}{s} + \frac{2}{s^{5/2}} - \frac{1}{s^2}$

$\Rightarrow \frac{2t^{1/2}}{\Gamma(\frac{1}{2}+1)} - 1 + \frac{2t^{3/2}}{\Gamma(\frac{3}{2}+1)} - \frac{1t^1}{\Gamma(2)}$

UNIT-
INVERSE LAPLACE TRANSFORM

① $L^{-1}\left\{\frac{K}{s}\right\} = K$

② $L^{-1}\left\{\frac{1}{s^2+a^2}\right\} = \frac{1}{a}\sin at$

③ $L^{-1}\left\{\frac{s}{s^2+a^2}\right\} = \cos at$

④ $L^{-1}\left\{\frac{1}{s^2-a^2}\right\} = \frac{1}{a}\sinh at$

⑤ $L^{-1}\left\{\frac{s}{s^2-a^2}\right\} = \cosh at$

⑥ $L^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$

⑦ $L^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}$

⑧ $L^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{\lceil(n+1)}$

↑

Inverse Transform of elementary function

Q. Find $f(t)$ for the following functions

① $\frac{3}{s+4} \quad \frac{-1}{s-1}$

$\Rightarrow 3e^{-4t} - e^{t}$

② $\frac{8s}{s^2+16}$

$\Rightarrow 8\cos 4t$

③ $\frac{1}{2s-5} = \frac{1}{2(s-5/2)}$

$\Rightarrow \frac{1}{2} e^{5/2 t}$

④ $\frac{6}{s^2-4}$

$\Rightarrow \frac{6}{2}\sinh 2t = 3\sinh 2t$

⑤ $\frac{3s-12}{s^2+8} = \frac{3s}{s^2+8} - \frac{12}{s^2+8}$

$\Rightarrow 3\cos\sqrt{8}t - 12\frac{\sin\sqrt{8}t}{\sqrt{8}}$

⑥ $s^{-7/2}$

$\Rightarrow \frac{1}{s^{7/2}} = \frac{t^{7/2}}{\lceil(7/2+1)} = \frac{8\,t^{5/2}}{15\sqrt{\pi}}$

$= \frac{t^{7/2}}{\frac{7}{2}\times\frac{5}{2}\times\frac{3}{2}\times\frac{1}{2}\sqrt{\pi}}$

$= \frac{15\,t^{5/2}}{8\sqrt{\pi}}$

Q Find the Laplace Transform of the following functions

(1) $L\{t^2 \delta(t-2)\}$

(2) $L\{\sin t\, \delta(t-\pi/2)\}$

(3) $L\{t^4 \delta(t-3)\}$

(4) $L\left\{\frac{\delta(t-a)}{t}\right\}$

(5) $L\{(t-1)^2 \delta(t-a)\}$

(6) $L\left\{\frac{2\delta(t-3)+3\delta(t-2)}{t}\right\}$

(7) $L\{\cosh 3t\, \delta(t-2)\}$

(1) $L\{t^2\delta(t-2)\} = e^{-2s} f(2) = e^{-2s} 4$

Here $f(t) = t^2$

$f(2) = 4$

(2) $L\{\sin t\, \delta(t-\pi/2)\} = e^{-\pi/2 s} F(\pi/2) = e^{-\pi/2 s}(1)$

Here $f(t) = \sin t$

$f(\pi/2) = 1$

(3) $L\{t^4 \delta(t-3)\} = e^{-3s} F(3) = e^{-3s}(3)^4$

Here $f(t) = t^4$

$f(3) = (3)^4$

(4) $L\left\{\frac{\delta(t-a)}{t}\right\} = e^{-as} \times \frac{1}{a}$

UNIT IMPULSE FUNCTION / DIRAC DELTA FUNCTION

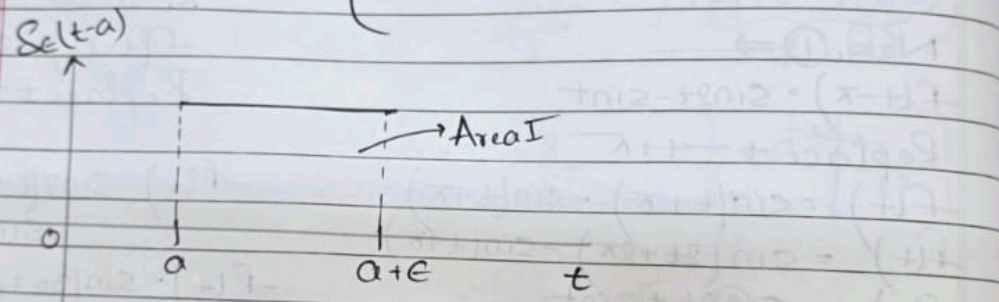
$$\delta(t-a) = \lim_{\epsilon \to 0} \delta_\epsilon(t-a), \quad a \geq 0$$

$$\delta_\epsilon(t-a) = \begin{cases} \frac{1}{\epsilon} & a < t < a+\epsilon \\ 0 & \text{otherwise} \end{cases}$$

↳ Unit impulse function is denoted by

$$\delta(t-a) = \lim_{\epsilon \to 0} \delta_\epsilon(t-a)$$

where, $\delta_\epsilon(t-a) = \begin{cases} \frac{1}{\epsilon} & a < t < a+\epsilon \\ 0 & \text{otherwise} \end{cases}$



As $\epsilon \to 0$ (width of the strip becomes smaller) + the height of the strip increases indefinitely in such a way that area remains unity.

$$\boxed{L\{f(t)\delta_\epsilon(t-a)\} = e^{-as} f(a)}$$

(c) $f(t) = \begin{cases} t-1 & 1 < t \le 2 \\ 3-t & 2 < t \le 3 \\ 0 & t > 3 \end{cases}$

Sol:

$f(t) = f_1(t) + [f_2(t) - f_1(t)]U(t-a) + [f_3(t) - f_2(t)]U(t-b)$

$f(t) = (t-1) + [(3-t) - (t-1)]U(t-2) + [0 - (3-t)]U(t-3)$

$f(t) = (t-1) + [3-t-t+1]U(t-2) + [t-3]U(t-3)$

$f(t) = (t-1) + [(4-2t)U(t-2)] + [(t-3)U(t-3)]$

Apply L.T on both sides

$L\{f(t)\} = L\{(t-1)\} + \underbrace{L\{(4-2t)U(t-2)\}}_{①} + \underbrace{L\{(t-3)U(t-3)\}}_{②}$

$L\{f(t)\} = \frac{1}{s^2} - \frac{1}{s} + e^{-2s}\left[\quad -\frac{2}{s^2} \quad\right] + e^{-3s}\left[\frac{1}{s^2}\right]$

Now ① ⇒

$f(t-2) = 4-2t$

Replace $t \to t+2$

$f(t) = 4 - 2(t+2)$

$= 4 - 2t - 4$

$f(t) = -2t$

$\boxed{L\{f(t)\} = \frac{-2}{s^2}}$

Now ② ⇒

$f(t-3) = t-3$

Replace $t \to t+3$

$f(t) = t+3-3$

$\boxed{L\{f(t)\} = \frac{1}{s^2}}$

(15) $$f(t)=\begin{cases}\sin t & t<\pi \\ 1 & \pi<t\le 2\pi \\ \cos t & t>2\pi\end{cases}$$

Sol:

$$f(t)=f_1(t)+[f_2(t)-f_1(t)]u(t-a)+[f_3(t)-f_2(t)]u(t-b)$$

$$f(t)=\sin t+[1-\sin t]u(t-\pi)+[\cos t-1]u(t-2\pi)$$

$$f(t)=\sin t+\{(1-\sin t)u(t-\pi)\}+\{(\cos t-1)u(t-2\pi)\}$$

Apply LT on both sides

$$L\{f(t)\}=L\{\sin t\}+\underbrace{L\{(1-\sin t)u(t-\pi)\}}_{①}+\underbrace{L\{(\cos t-1)u(t-2\pi)\}}_{②}$$

$$=\frac{1}{s^2+1}+e^{-\pi s}\left[\frac{1}{s}+\frac{1}{s^2+1}\right]+e^{-2\pi s}\left[\frac{s}{s^2+1}-\frac{1}{s}\right]$$

Now ① ⇒

$f(t-\pi)=1-\sin t$

Replace $t\to t+\pi$

$f(t)=1-\sin(t+\pi)$

$f(t)=1+\sin t$

$$\boxed{L\{f(t)\}=\frac{1}{s}+\frac{1}{s^2+1}}$$

Now ② ⇒

$f(t-2\pi)=\cos t-1$

Replace $t\to t+2\pi$

$f(t)=\cos(t+2\pi)-1$

$=\cos t-1$

$$\boxed{L\{f(t)\}=\frac{s}{s^2+1}-\frac{1}{s}}$$

Now, $f(t-1) = 1-e^t$

Replace $t \to t+1$

$f(t) = 1-e^{(t+1)}$

$= 1-e^t e^1$

$L\{f(t)\} = \frac{1}{s} - \frac{e^1}{(s-1)} = F(s)$

③ $f(t) = \begin{cases} \pi - t & 0<t<\pi \\ \sin t & t>\pi \end{cases}$

Sol: $f_2(t) = f_1(t) + [f_2(t) - f_1(t)]\, U(t-a)$

$f(t) = (\pi - t) + [\sin t - (\pi - t)]\, U(t-\pi)$

$f(t) = (\pi - t) + (\sin t - \pi + t)\, U(t-\pi)$

Apply LT on both sides

$L\{f(t)\} = L\{\pi - t\} + L\{(\sin t - \pi + t)\, U(t-\pi)\}$

$= \frac{\pi}{s} - \frac{1}{s^2} + e^{-\pi s}\left[\frac{-1}{s^2+1} + \frac{1}{s^2}\right]$

Now,

$f(t-\pi) = \sin t - \pi + t$

Replace $t \to t+\pi$

$f(t) = \sin(t+\pi) - \pi + t + \pi$

$f(t) = -\sin t + t$

$L\{f(t)\} = -\frac{1}{s^2+1} + \frac{1}{s^2}$

NOTE:

① If $f(t) = \begin{cases} f_1(t) & 0<t<a \quad t\leq a \\ f_2(t) & t>a \end{cases}$

$$f(t) = f_1(t)\left[u(t-0) - u(t-a)\right] + f_2(t)\,u(t-a)$$
$$= f_1(t)\left[1 - u(t-a)\right] + f_2(t)\,u(t-a)$$
$$= f_1(t) - f_1(t)\,u(t-a) + f_2(t)\,u(t-a)$$

② $f(t) = \begin{cases} f_1(t) & 0<t<a \\ f_2(t) & 0<t<b \\ f_3(t) & t>b \end{cases}$

$$f(t) = f_1(t) + \left[f_2(t) - f_1(t)\right]u(t-a) + \left[f_3(t) - f_2(t)\right]u(t-b)$$

Q Express the following function in terms of unit step function and hence find the Laplace Transform

Sol:

① $f(t) = \begin{cases} t & 0<t<2 \\ t^2 & t>2 \end{cases}$

Sol: $f(t) = \begin{cases} t = f_1(t) & 0<t<2 \\ t^2 = f_2(t) & t>2 \end{cases}$

$$f(t) = f_1(t) + \left[f_2(t) - f_1(t)\right]u(t-a)$$
$$f(t) = t + \left[(t^2 - t)\,u(t-2)\right]$$

(n) $te^{-t}u(t-2)$

Sol: $f(t-2) = te^{-t}$

Replace $t \to t+2$

$f(t) = (t+2)e^{-(t+2)}$

$= (t+2)e^{-t}e^{-2}$

$= e^{-2}[te^{-t} + 2e^{-t}]$

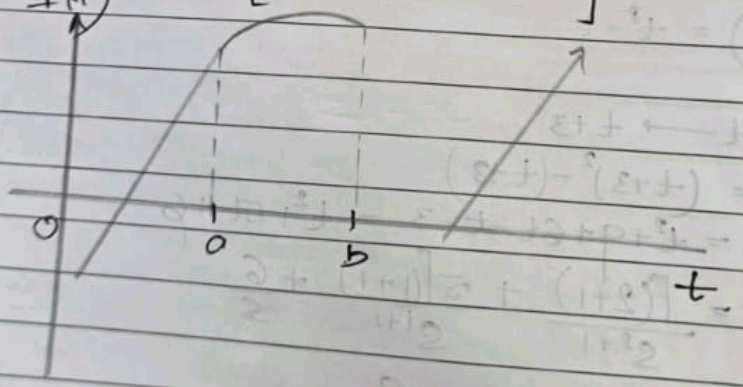
$L\{f(t)\} = e^{-2}\left[\frac{\lceil(1+1)}{s^{1+1}}e^{-t} + \frac{2}{s}e^{-t}\right]$

$= e^{-2}\left[\frac{1}{(s+1)^2} + \frac{2}{(s+1)}\right]$

$\therefore L\{te^{-t}u(t-2)\} = e^{-2s}e^{-2}\left[\frac{1}{(s+1)^2} + \frac{2}{(s+1)}\right]$

$\Rightarrow$ If $f(t)\begin{cases}\phi(t) & a<t<b \\ 0 & \text{Otherwise}\end{cases}$

$f(t) = \phi(t)[u(t-a) - u(t-b)]$



(3) $e^{-t}\{\sin t\, u(t-\pi/2)$

Sol: Given:

$f(t) - f(t-\pi/2) = e^{-t}\sin t$

Replace $t \longrightarrow t+\pi/2$

$f(t) = e^{-(t+\pi/2)}\sin(t+\pi/2)$

$f(t) = e^{-t}e^{-\pi/2}\cos t$

$f(t) = e^{-\pi/2}e^{-t}\cos t$

$L\{f(t)\} = e^{-\pi/2}\dfrac{(s+1)}{(s+1)^2+1} = F(s)$

$\therefore L\{e^{-t}\sin t\, u(t-\pi/2)\} = e^{-\pi/2 s}\cdot e^{-\pi/2}\left(\dfrac{s+1}{(s+1)^2+1}\right)$

(4) $(t^2-t)u(t-3)$

Sol: Given:

$f(t-3) = t^2-t$

Replace $t \longrightarrow t+3$

$f(t) = (t+3)^2-(t+3)$

$= t^2+9+6t-t-3 = t^2+5t+6$

$L\{f(t)\} = \dfrac{\lfloor(2+1)}{s^{2+1}} + 5\dfrac{\lfloor(1+1)}{s^{1+1}} + \dfrac{6}{s}$

$= \dfrac{2!}{s^3} + \dfrac{5}{s^2} + \dfrac{6}{s}$

$\therefore L\{(t^2-t)u(t-3)\} = e^{-3s}\left(\dfrac{2!}{s^3}+\dfrac{5}{s^2}+\dfrac{6}{s}\right)$

NOTE: $L\{\cos A$

Q $L\left\{\left(\frac{\sin 2t}{\sqrt{t}}\right)^2\right\}$

Sol: $L\left\{\frac{\sin^2 2t}{t}\right\}$

$f(t) = \sin^2 2t$

$\therefore \sin^2 2t = \frac{1-\cos 4t}{2}$

$$L\{f(t)\} = \frac{1}{2}\left[\frac{1}{s} - \frac{s}{s^2+16}\right]$$

WKT,

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s)\,ds$$

$$L\left\{\left(\frac{\sin 2t}{\sqrt{t}}\right)^2\right\} = \int_s^\infty \frac{1}{2}\left[\frac{1}{s} - \frac{s}{s^2+16}\right] ds$$

$$= \frac{1}{2}\left[\log s - \frac{1}{2}\log(s^2+16)\right]_s^\infty$$

$$= \frac{1}{2}\left[\log s - \log\sqrt{s^2+16}\right]_s^\infty$$

$$= \frac{1}{2}\left[\log \frac{s}{\sqrt{s^2+16}}\right]_s^\infty$$

$$= \frac{1}{2}\left[\log \frac{1}{\left(\sqrt{1+16/s^2}\right)}\right]_s^\infty$$

$$= \frac{1}{2}\left[0 - \log \frac{s}{\sqrt{s^2+16}}\right]$$

$$= \frac{-1}{2}\log \frac{s}{\sqrt{s^2+16}} //$$

$$L\left\{\frac{\sin t \sin 5t}{t}\right\} = \int_s^\infty \frac{1}{2}\left[\frac{s}{s^2+16} - \frac{s}{s^2+36}\right] ds$$

$$= \frac{1}{2}\left[\frac{1}{2}\log(s^2+16) - \frac{1}{2}\log(s^2+36)\right]_s^\infty$$

$$= \frac{1}{4}\left[\log\frac{(s^2+16)}{(s^2+36)}\right]_s^\infty$$

$$= \frac{1}{4}\left[\log\frac{(1+16/s^2)}{(1+36/s^2)}\right]_s^\infty$$

$$= \frac{1}{4}\left[0 - \log\frac{(1+16/s^2)}{(1+36/s^2)}\right]$$

$$= \frac{1}{4}\left[-\log\frac{(s^2+16)}{(s^2+36)}\right] = -\frac{1}{4}\log\frac{(s^2+16)}{(s^2+36)}$$

Now, $L\left\{e^{2t}\frac{\sin t \sin 5t}{t}\right\}$

$$= -\frac{1}{4}\log\frac{((s-2)^2+16)}{((s-2)^2+36)}$$

Q $L\left\{\frac{\sin^3 t}{t}\right\}$, Hence find $L\left\{e^{2t}\frac{\sin^3 t}{t}\right\}$

Sol $\sin^3 t = \frac{3\sin t - \sin 3t}{4}$

$f(t) = \sin^3 t$

$$L\{f(t)\} = \frac{1}{4}\left[\frac{3}{s^2+1} - \frac{3}{s^2+9}\right]$$

$$= \frac{3}{4}\left[\frac{1}{s^2+1} - \frac{1}{s^2+9}\right]$$

Q $L\left\{\frac{\sin t}{t}\right\}$

Sol Let $f(t) = \sin t$

$L\{f(t)\} = \frac{1}{s^2+1} = F(s)$

WKT $L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s)\,ds$

$L\left\{\frac{\sin t}{t}\right\} = \int_s^\infty \frac{1}{s^2+1}\,ds$

$= \left[\tan^{-1}(s)\right]_s^\infty = \tan^{-1}(\infty) - \tan^{-1}(s)$

$= \pi/2 - \tan^{-1}(s) = \cot^{-1}(s)$

Q $L\left\{\frac{e^t \sin t}{t}\right\}$

↳ then $L\left\{\frac{e^t \sin t}{t}\right\} = \cot^{-1}(s-1)$

Q $L\left\{\frac{\sin t \sin 5t}{t}\right\}$ hence find $L\left\{\frac{e^{2t} \sin t \sin 5t}{t}\right\}$

Sol $\sin 5t \sin t = \frac{-1}{2}\left[\cos 6t - \cos 4t\right]$

$= \frac{1}{2}\left[\cos 4t - \cos 6t\right]$

Let $f(t) = \sin t \sin 5t$

$L\{f(t)\} = \frac{1}{2}\left[\frac{s}{s^2+16} - \frac{s}{s^2+36}\right]$

Now,

$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s)\,ds$

Q Find $L\left\{\frac{e^{at}-\cos bt}{t}\right\}$

Sol: Let $f(t) = e^{at} - \cos bt$

$L\{f(t)\} = \frac{1}{(s-a)} - \frac{s}{s^2+b^2} = F(s)$

Now, WKT

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s)\,ds$$

$$L\left\{\frac{e^{at}-\cos bt}{t}\right\} = \int_s^\infty \frac{1}{(s-a)} - \frac{s}{s^2+b^2}\,ds$$

$$= \left[\log(s-a) - \frac{1}{2}\log(s^2+b^2)\right]_s^\infty$$

$$= \left[\log(s-a) - \log\sqrt{(s^2+b^2)}\right]_s^\infty$$

$$= \left[\log\frac{(s-a)}{\sqrt{s^2+b^2}}\right]_s^\infty$$

$$= \left[\log\frac{s(1-a/s)}{s\sqrt{1+b^2/s^2}}\right]_s^\infty$$

$$= \left[0 - \log\frac{(1-a/s)}{\sqrt{1+b^2/s^2}}\right]$$

$$= -\log\left[\frac{\frac{s-a}{s}}{\sqrt{\frac{s^2+b^2}{s^2}}}\right] = -\log\frac{(s-a)}{\sqrt{s^2+b^2}}$$

$$\log\frac{\sqrt{s^2+b^2}}{(s-a)}$$

① Sol: $L\{t^{5/2}\sin^2 2t\}$

Let $f(t) = t^{5/2}$

$L\{f(t)\} = \frac{\lfloor(5/2+1)}{s^{5/2+1}} = \frac{\frac{5}{2}\times\frac{3}{2}\times\frac{1}{2}\lfloor\frac{1}{2}}{s^{7/2}}$

$= \frac{15\sqrt{\pi}}{8\,s^{7/2}} = F(s)$

$\sin^2 2t = \left(\frac{e^{2t}-e^{-2t}}{2}\right)^2$

$= \left[\frac{e^{4t}+e^{-4t}-2}{4}\right]$

$L\{t^{5/2}\sin^2 2t\} = \left[\frac{e^{4t}+e^{-4t}-2}{4}\right]\frac{15\sqrt{\pi}}{8\,s^{7/2}}$

$= \frac{15}{32}\left[\frac{\sqrt{\pi}}{(s-4)^{7/2}} + \frac{\sqrt{\pi}}{(s+4)^{7/2}} - \frac{2}{s^{7/2}}\right]$

② $L\{t\sin 3t\}$

Sol: Let $f(t) = \sin 3t$

$L\{f(t)\} = \frac{3}{s^2+9}$

$\because L\{t\sin 3t\} = (-1)\frac{d}{ds}\frac{3}{s^2+9}$

$= -3\left[\frac{-1}{(s^2+9)^2}\times 2s\right]$

$= \frac{6s}{(s^2+9)^2}$

$$= \frac{1}{2}\left[\frac{2}{s^3} + \left[\frac{2s(s^2+36) - 4s(s^2-36)}{(s^2+36)^3}\right]\right]$$

$$= \frac{1}{2}\left[\frac{2}{s^3} + \left[\frac{2s^3+72s-4s^3+144s}{(s^2+36)^3}\right]\right]$$

$$= \frac{1}{2}\left[\frac{2}{s^3} + \left[\frac{216s-2s^3}{(s^2+36)^3}\right]\right]$$

$$= \frac{1}{s^3} + \frac{108s-s^3}{(s^2+36)^3}$$

Q The functional form of signal in time domain is given by $\{t^2 e^{3t} \sin 3t\}$, find its transform in frequency domain.

Sol: Consider,

$$f(t) = e^{3t}\sin 3t$$

$$L\{f(t)\} = \frac{3}{s^2+9} = F(s)$$

Now,

$$L\{t^2 \sin 3t\} = (-1)^2 \frac{d^2}{ds^2}\left(\frac{3}{s^2+9}\right)$$

$$= \frac{d}{ds}\frac{d}{ds}\left(\frac{3}{s^2+9}\right)$$

$$= 3\frac{d}{ds}\left(\frac{-1\times 2s}{(s^2+9)^2}\right)$$

$$= -6\frac{d}{ds}\left[\frac{s}{(s^2+9)^2}\right]$$

$$= -6\left[\frac{(s^2+9)^2 - s\times 2(s^2+9)\times 2s}{(s^2+9)^4}\right]$$

$$= -6\left[\frac{(s^2+9) - 4s^2}{(s^2+9)^3}\right]$$

$$= -6\left[\frac{9-3s^2}{(s^2+9)^3}\right]$$